Entropic Scalar EFT

Microstructure, Gravity, and the Retained Quantum Sector
Compact Derivation

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Abstract

Spacetime is proposed as the long-wavelength description of finite local entanglement capacity. Matter commits that capacity, mass measures the commitment, and weak gravity measures its fractional depletion.

Positive oriented matching gives a seven-state channel and nine-state record; single-copy routing gives 1680 boundary states. Their closure spectrum fixes η_* and $g_{\mathrm{share,eff}}$. Faithful renewal and a selected marked generator give an electron-calibrated cadence, $L_* = c\tau_*$ and $G_* = c^3 L_*^2/\hbar$. Coherent bonds supply $\ln 2$ per cut. Angular transfer and a Poisson/longest-chain volume calibration give $G_0/G_* = 0.1055193 m_4^{8/3}$, approximately 1.03 at the illustrative $m_4 = 2.35$.

Connected primitive patches realize normalized causal-triangulation histories. Coarse areas reconstruct a metric; selected DeWitt dynamics gives the Einstein quadratic response and a positive homogeneous physical inner product. A finite geometry–orbital operator preserves the native mass gap, clock, and encoded evolution under refinement. Source-aware reduction, positive reconstruction, and Gaussian elimination connect these constructions. A preregistered connected $\alpha = 7$ test gives $G_{\mathrm{vol}}/G_0 = 1.064$ with logarithmic standard error 0.089. Start independence, a universal propagation cone, and a common interacting Newton coefficient remain open.

The matched Einstein branch gives the nonlinear static lapse equation, Newton's law, no slip, and GR post-Newtonian values. The shell model correlates charged-lepton masses. The transverse EFT gives $a_0 = [g_{\mathrm{share,eff}}/(4\pi^2)]cH_0$ and the spherical exponential radial-acceleration law. Its director completion passes development comparisons and two old-disk checks; a conditional live-stock model retains substantial halo tilt for eight Gyr. Ensemble-clock release with composition-preserving allocation gives exponential depletion and a local exhaustion boundary. Direct geodesic fits of linear depletion and exponential allocation differ from ΛCDM by less than one in χ^2.

1. Inputs, Claims, and Logical Grades

The derivation begins only after five theory-defining inputs are fixed:

1. **Finite local capacity.** The vacuum has a bounded local budget of entanglement.
2. **Information–geometry equivalence.** Geometry is the continuum description of that substrate; in the weak field the potential is the fractional deficit of available capacity.
3. **Mass–entropy equivalence.** Matter is localized committed capacity and

$$m(\ell) = \kappa_m(\ell)\,\Delta S.$$

   For an elementary fermionic exclusion, $\Delta S_f = \ln 2$.
4. **Many-Pasts.** One realized present is supported by the decoherent past histories compatible with its physical records. Laboratory probabilities remain the ordinary Born probabilities of the retained quantum sector.

5. **Tetrahedral UV architecture.** The minimal three-dimensional volumetric cell is tetrahedral; a shared face begins with two primitive half-integer slots, compared by positive oriented matching. Four relational ports draw from one single-copy channel family, and the two global orientations are distinct.

Faithful full-support resolution maximizes joint entropy at fixed admissibility marginals. Its temporal application is maximum caliber; finite capacity alone does not imply it. The selected realization specifies the closure coordinates, marked routing, determinant-survival action, additive charged generator, shell encoding, and invariant-speed conversion. Its spatial completion adds a coherent bond state, absolute angular transfer, event-to-volume calibration, normalized geometric packets, and choices of metric dynamics and orbital transport. Each choice enters the derivation where it is used. The broad commitments admit other realizations.

Four grades track the logical status of each result. *Closed* means derived within the explicitly named inputs of the relevant branch. *Fixed* means no continuous phenomenological freedom remains once a named branch is adopted. *Conditional* means the conclusion follows from a stated matching or completion premise. *Open* means the construction is specified but a necessary microscopic, continuum, or empirical step remains.

The dependency spine is

$$\begin{gathered}
\text{selected tetrahedral vertex} \longrightarrow 7+9 \longrightarrow 1680 \longrightarrow \eta_*, g_{\text{share,eff}}, \\
\text{faithful renewal + selected charged action + electron} \longrightarrow \tau_*, \qquad L_* = c\tau_*, \\
G_* := c^3 L_*^2/\hbar, \qquad \text{source map + physical matching} \longrightarrow G = G_*, \\
\text{bond state + angular transfer + event calibration} \longrightarrow G_0/G_* = 0.1055193\, m_4^{8/3}, \\
\text{Einstein parent + static capacity coordinate} \longrightarrow \Phi = \Psi, \quad \mathrm{PPN}_{\mathrm{GR}}.
\end{gathered}$$

The galactic, cluster, cosmological, and dark-energy sectors add their own premises. Covariant formulas use signature $(-,+,+,+)$ and $x^0 = ct$ unless $c = 1$ is explicitly stated. The causal scale L_* does not by itself determine a graph bond length or a physical tetrahedron's volume.

2. Why the UV Pair Has Seven Transmitted States

Two cells sharing one face contribute

$$\mathcal{H}_f = V_{j_0} \otimes V_{j_0} = \bigoplus_{J=0}^{2j_0} V_J.$$

Orientation reversal uses the dual spin generators,

$$\widetilde{\mathbf{J}}_R = \mathrm{Ad}_{g_{LR}}\, \mathbf{J}_R^{\vee}, \qquad J_R^{\vee a} = -(J_R^a)^T.$$

These obey the same $\mathfrak{su}(2)$ algebra. After transporting the right slot into the left orientation, exact positive coherent matching is the direction-averaged Gram operator

$$\mathcal{G}_{\text{sharp}} = (4j_0+1) \int \frac{d\Omega_{\mathbf{n}}}{4\pi} |j_0, \mathbf{n}; j_0, \mathbf{n}\rangle\langle j_0, \mathbf{n}; j_0, \mathbf{n}|.$$

The coherent-state resolution of the identity on the maximal coupled irrep gives

$$\boxed{\mathcal{G}_{\text{sharp}} = P_{2j_0}}.$$

Thus oriented matching transmits V_{2j_0}, not an independently chosen alphabet.

This sharp result is the endpoint of a positive finite-width family rather than an isolated projector. Subtracting the irreducible aligned-spin mismatch floor defines

$$Q_f = \frac{1}{2}\left[(\mathbf{J}_L - \widetilde{\mathbf{J}}_R)^2 - 2j_0 I\right].$$

On the coupled irrep V_J,

$$Q_f\big|_{V_J} = \frac{1}{2}\left[(2j_0)(2j_0+1) - J(J+1)\right] I_{V_J},$$

so $Q_f \geq 0$ and $\ker Q_f = V_{2j_0}$. The positive transfer

$$T_w = e^{-wQ_f}, \qquad w > 0,$$

therefore converges monotonically to P_{2j_0}. At the branch selected below, $j_0 = 3/2$,

$$\boxed{Q_f = 6P_0 + 5P_1 + 3P_2}, \qquad \boxed{T_w = e^{-6w}P_0 + e^{-5w}P_1 + e^{-3w}P_2 + P_3}.$$

A minimal homogeneous field action on the complete pair space is

$$\Gamma_f = \int d^dx \left[Z|\partial\Phi|^2 + r\Phi^\dagger\Phi + \kappa\Phi^\dagger Q_f \Phi + \frac{u}{2}(\Phi^\dagger\Phi)^2\right], \qquad Z, \kappa, u > 0.$$

Writing ρ_J for the occupation of V_J, its potential is

$$V = r\rho + \frac{u}{2}\rho^2 + 6\kappa\rho_0 + 5\kappa\rho_1 + 3\kappa\rho_2.$$

For $r < 0$ the unique support of the homogeneous minimum is

$$\boxed{\rho_3 = -\frac{r}{u}, \qquad \rho_0 = \rho_1 = \rho_2 = 0},$$

and the orthogonal fluctuations have positive relative masses

$$\boxed{m^2_{0|3} : m^2_{1|3} : m^2_{2|3} = 6 : 5 : 3}.$$

Exact coherent matching selects the maximal channel, which is also the unique homogeneous phase of the minimal positive mismatch action. A more fundamental host vertex must reproduce this ordering or pass the corresponding reduced-Hessian inequalities.

The retained marked response consists of one present and one history vector strand. In three spatial dimensions a vector is V_1, so

$$\mathcal{H}_{\text{mark}} = V_1^P \otimes V_1^H = V_0 \oplus V_1 \oplus V_2, \qquad \dim \mathcal{H}_{\text{mark}} = 9.$$

Maximal fusion of two primitive V_j slots leaves the complement

$$Q(j) = (V_j \otimes V_j) \ominus V_{2j} = \bigoplus_{J=0}^{2j-1} V_J.$$

Both decompositions are multiplicity-free, hence

$$\mathcal{H}_{\text{mark}}(s) \cong Q(j) \iff 2s = 2j - 1 \iff \boxed{j = s + \frac{1}{2}}.$$

With the independently fixed spatial-vector response $s = 1$,

$$\boxed{j_0 = \frac{3}{2}}, \qquad \boxed{V_{2j_0} = V_3}, \qquad \boxed{|M| = 7}.$$

The full pair space has dimension $16 = 7 + 9$. In the selected sharp reversible vertex, unitarity fixes the retained output to the orthogonal complement:

$$U_{\text{sharp}}|\psi\rangle = F_3 P_3|\psi\rangle|0\rangle_R + |\varnothing\rangle L|\psi\rangle,$$

with

$$L^\dagger L = P_0 + P_1 + P_2, \qquad U_{\rm sharp}^\dagger U_{\rm sharp} = P_3 + P_0 + P_1 + P_2 = I_{16}.$$

Therefore the seven-state link and nine-state marked record are complementary outputs of one reversible pair map. The hard-core flags and detailed routing are selected microscopic structure of the displayed vertex; the $7 + 9$ carrier incidence is forced inside that vertex.

The equal normalization of the retained nine-state carrier also removes the finite matching width in the minimal reversible dilation. At finite w, the complementary squared amplitudes on the nonmaximal blocks are

$$\ell_0^2 = 1 - e^{-6w}, \qquad \ell_1^2 = 1 - e^{-5w}, \qquad \ell_2^2 = 1 - e^{-3w}.$$

The marked vertex has one common Gram normalization,

$$G_{\rm mark} = I_9 = P_0 + P_1 + P_2.$$

Without inserting compensating J-dependent couplings, exact identification of this marked carrier with the canonical environment therefore requires

$$\boxed{w \to \infty, \qquad T_w \to P_3, \qquad I - T_w \to P_0 + P_1 + P_2}.$$

Thus the sharp $7 + 9$ split is selected within the minimal equal-weight reversible incidence; a finite physical width would define a different microscopic vertex with additional block-dependent structure.

The four faces are distinguishable ports drawing from one single-copy fermionic channel family, $n_m \in \{0, 1\}$. Two ports cannot display the same m because that would require $n_m = 2$. Hence the face assignment is injective and ordered. Including the two global orientations,

$$\boxed{\Omega_{\rm tet} = 2P(7, 4) = 1680}, \qquad \boxed{g_{\rm share,max} = \ln 1680 = 7.42654907240}.$$

Full use of the fixed seven-letter alphabet also forces the unconditioned prior to be uniform, because $H(\mu) \le \ln 7$ with equality only for $\mu(m) = 1/7$.

Resulting branch constraints. Within the selected sharp reversible incidence, the chain

$$d = 3 \to V_1 \to j_0 = \tfrac{3}{2} \to V_3 \to 7 \to 1680$$

contains no continuous parameter and uses no smallest-spin axiom. A different microscopic vertex would define a different discrete branch; the nearby branch audit is therefore a model comparison, not a continuous fit.

3. Exact Admissibility Closure and the Effective Entropy

For a boundary state $b = (m_1, \ldots, m_4)$ with $m_i \in \{-3, \ldots, 3\}$, define

$$S = \sum_i m_i, \qquad \Sigma^2 = \sum_i m_i^2.$$

The tetrahedral normal frame obeys $\hat{n}_i \cdot \hat{n}_j = -1/3$ for $i \neq j$. For the product of four $j = 3$ face states, the quantum expectation of the squared closure operator is

$$\boxed{K^2(b) = 48 - \frac{1}{3}(S^2 - \Sigma^2)} = |\mathbf{c}(b)|^2 + \sum_i \left[j(j+1) - m_i^2 \right].$$

Here $\mathbf{c}(b) = \sum_i m_i \hat{n}_i$. This is a *soft admissibility* invariant, not the exact Gauss constraint.

The maximum-entropy family at fixed quadratic closure information is

$$p_\eta(b) = \frac{e^{-\eta K^2(b)}}{Z(\eta)}, \qquad Z(\eta) = \sum_b e^{-\eta K^2(b)}.$$

Because closure failure has three independent components, normalized isotropic evidence carries determinant factor $\eta^{3/2}$:

$$\mathcal{F}(\eta) = \ln Z(\eta) + \frac{3}{2}\ln \eta.$$

Hence

$$\mathcal{F}'(\eta) = 0 \Longleftrightarrow \boxed{\langle K^2\rangle_\eta = \frac{3}{2\eta}}.$$

The finite spectrum is

K^2	122/3	134/3	142/3	146/3	152/3	154/3
n	96	96	96	288	192	144

K^2	158/3	54	164/3	166/3	170/3
n	384	192	48	96	48

with total multiplicity 1680. Therefore

$$Z(\eta) = \sum_a n_a e^{-\eta K_a^2}, \qquad \langle K^2\rangle_\eta = \frac{\sum_a n_a K_a^2 e^{-\eta K_a^2}}{\sum_a n_a e^{-\eta K_a^2}}.$$

Global uniqueness follows from $x_{\min} = 122/3$ and $x_{\max} = 170/3$. Every stationary point lies in

$$\mathcal{I} = \left[\frac{3}{2x_{\max}}, \frac{3}{2x_{\min}}\right],$$

because $\mathcal{F}'$ is positive below and negative above this interval. The bounded-spectrum variance obeys $\mathrm{Var}(K^2) \le 64$, so throughout $\mathcal{I}$,

$$\mathcal{F}'' \le 64 - \frac{2}{3}x_{\min}^2 = -1038.5185\ldots < 0.$$

Since $\mathcal{F} \to -\infty$ at both endpoints of $\eta \in (0,\infty)$, there is exactly one global maximum,

$$\boxed{\eta_* = 0.0298668443935}.$$

At that point

$$\mathrm{Var}_{\eta_*}(K^2) = 15.6889750, \qquad \mathcal{F}''(\eta_*) = 15.69 - \frac{3}{2\eta_*^2} = -1665.9 < 0,$$

consistent with the global uniqueness proof. The effective sharing entropy is then fixed:

$$\boxed{g_{\text{share,eff}} = -\sum_b p_{\eta_*}(b)\ln p_{\eta_*}(b) = 7.41980002357.}$$

The identity

$$\eta_*\langle K^2\rangle_{\eta_*} = \frac{3}{2}$$

will also supply the reciprocal transverse factor 2/3 used downstream.

Parameter status. Once the seven-state tetrahedral ensemble, the quadratic closure operator, and normalized three-component evidence are fixed, η_* and $g_{\text{share,eff}}$ are evaluations of a finite spectrum. No phenomenological coefficient remains in this step.

4. Faithful Histories Force Renewal

The closure calculation fixes the one-time marginal $p_* = p_{\eta_*}$. For any joint law P_n with those marginals,

$$D_{\mathrm{KL}}(P_n \| p_*^{\otimes n}) = nH(p_*) - H(P_n) \geq 0,$$

with equality only for the product law. Here D_{KL} is the Kullback–Leibler divergence. Applying faithful resolution to every finite temporal block therefore selects independent records. Equivalently, faithful full-support resolution applied to paths requires maximal entropy rate. For any stationary process,

$$h_\mu = \lim_{n\to\infty} H(B_0|B_{-1}, \ldots, B_{-n}) \leq H(B_0) = g_{\mathrm{share,eff}}.$$

Equality holds only if B_0 is independent of its entire past. Stationarity then makes the process i.i.d. with the same marginal at every step. For a stationary Markov process,

$$H(B_{t+1}|B_t) = H(B_{t+1}) - I(B_t; B_{t+1}) \leq g_{\mathrm{share,eff}},$$

with equality iff $I(B_t; B_{t+1}) = 0$. Therefore

$$\boxed{K_*(b, b') = p_{\eta_*}(b')}.$$

Finite capacity by itself does not imply renewal; faithful full-support resolution does.

For seven channel layers with fixed marginal entropy $g_{\mathrm{share,eff}}$,

$$H(B_{-3}, \ldots, B_3) = 7g_{\mathrm{share,eff}} - \Delta, \qquad \Delta \geq 0.$$

Within the adopted support-to-mass map, correlation reduces support and raises the recurrence mass, $m \propto e^{\Delta}$. Fermionic exclusion caps the occupied channel count at seven. Thus the lightest resolved one-bit charged branch is

$$\boxed{k = 7, \qquad \Delta_7 = 0}.$$

The same no-memory endpoint is therefore selected both by maximum path entropy and by minimal recurrence mass.

5. Electron Scale from the Positive Marked-Transfer Spectrum

The state-weighted determinant of likelihood multiplication by p_{η_*} on one renewed layer is

$$\Delta_{\tau_p}(\mathsf{R}) = \exp\left(\sum_b p_b \ln p_b\right) = e^{-g_{\mathrm{share,eff}}}.$$

Across seven independent layers,

$$\boxed{r = e^{-7g_{\mathrm{share,eff}}}}.$$

This is a record-conditioned geometric transfer rate, not a collision probability of two independently sampled microstates. The selected determinant-survival action assigns the positive survival transfer

$$T^{(0)}_{\mathrm{surv}} = 1 - r,$$

and hence the exact raw gap

$$E^{(0)}_{\mathrm{raw}} = -\frac{\hbar}{\tau^{(0)}_*} \ln(1 - r).$$

The transverse export factor is 3/2. Before the finite marked correction, identifying the lightest charged gap with the electron gives

$$m_e c^2 = -\frac{3\hbar}{2\tau^{(0)}_*} \ln(1 - r),$$

so

$$\boxed{L_*^{(0)} = c\tau_*^{(0)} = -\frac{3}{2}\lambda_e \ln(1-r), \qquad \lambda_e = \frac{\hbar}{m_e c}.}$$

No measured gravitational quantity enters this calibration.

The selected marked decoration reuses the closure amplitude. To include its quantum variance in a real auxiliary vector, define

$$\mathbf{c}(b) = \sum_i m_i \hat{n}_i, \qquad \mathbf{C}(b) = \sqrt{\frac{K^2(b)}{|\mathbf{c}(b)|^2}}\,\mathbf{c}(b).$$

Injectivity gives $|\mathbf{c}|^2 = (4\Sigma^2 - S^2)/3 > 0$, so $C^2 = K^2$ everywhere. These are auxiliary response coordinates, not three commuting quantum-closure measurements. Tetrahedral symmetry gives

$$\langle C_a C_b \rangle = \frac{\langle K^2 \rangle}{3}\delta_{ab} = \frac{\delta_{ab}}{2\eta_*}.$$

The Gaussian identity

$$e^{-\eta_* C^2/2} = \int \frac{d^3\xi}{(2\pi)^{3/2}} \exp\left[-\frac{1}{2}\xi^2 + i\sqrt{\eta_*}\,\xi_a C_a\right]$$

makes the amplitude-level incidence $\sqrt{\eta_*}$. For each unordered occupied pair $e = (m, m')$, the two directed scalar returns have row operator

$$R_e = \frac{2}{7}\left(\langle m \to m'| + \langle m' \to m|\right),$$

so

$$B_e B_e^\dagger = 2\eta_* \left(\frac{2}{7}\right)^2 = \frac{8\eta_*}{49} \equiv u.$$

The canonical unitary dilation contributes $(1-u)^{1/2}$ per pair. Seven channels contain $\binom{7}{2} = 21$ unordered pairs. Present and history each carry three first excitations, giving nine marked states. Therefore

$$\boxed{\zeta_* = 9e^{-g_{\text{share,eff}}}\left(1 - \frac{8\eta_*}{49}\right)^{21/2} = 0.005123584484947.}$$

The finite transfer graph then gives

$$Z_\mu = 1 + \zeta_*, \qquad \boxed{Z_e = (1+\zeta_*)(1+7\zeta_*^2)}, \qquad Z_{\tau,2} = 1 + \frac{2}{7}\zeta_*.$$

The trace fixes motif weights, not their attachment to charged evolution. With $w = \zeta_*/9$, the electron weights $1, w, w^2, w^3$ have multiplicities $1, 9, 567, 5103$ and sum to Z_e. The selected additive-generator rule assigns each weight as an activity: if $a = -(3/2)\ln(1-r) > 0$ and n_e is the occupied-sector projector,

$$T(s+t) = T(s)T(t), \qquad T(s) = e^{-asn_e}, \qquad T_e = e^{-aZ_e n_e}.$$

Consequently $E_e = -(3\hbar/2\tau_*)Z_e \ln(1-r)$. The electron fixes τ_*; the continuum speed defines $L_* = c\tau_*$. The finite orbital construction in Section 14.6 preserves this rest gap and position coherence while coupling matter to changing geometry. Its common propagation cone across physical geometries and species, including radiative corrections, remains a continuum requirement. The calibrated scales are

$$\boxed{L_* = Z_e L_*^{(0)} = -\frac{3}{2}Z_e\lambda_e \ln\!\left(1 - e^{-7g_{\text{share,eff}}}\right) = 1.6162537024 \times 10^{-35}\ \text{m}.}$$

The induced gravitational scale is

$$\boxed{G_* := \frac{c^3 L_*^2}{\hbar} = \frac{9}{4}\frac{\hbar c}{m_e^2}Z_e^2 \ln^2\!\left(1 - e^{-7g_{\text{share,eff}}}\right) = 6.6742890813 \times 10^{-11}\ \text{SI},}$$

which is -0.073σ from CODATA 2022. The computation uses $\hbar = h/(2\pi)$ with exact SI h and c, and the CODATA electron anchor [1].

Finite alternative and mass minimum. Within the fixed seven-channel marginals, marked vertex, and selected logarithmic generator, the KL identity and convexity make the product readout the unique mass minimum. Every $k < 7$ competitor that retains the unit unmarked motif is heavier by at least a factor 1659.8886. A second fully positive finite construction changes the result by only 7.37×10^{-26} relatively, but that agreement does not select the generator: its attachment remains model data.

Historical status. The high-precision marked vertex was constructed after the residual in G and the lepton residuals were known. The numerical agreement is therefore a postdiction. Its nontrivial content is that one finite marked action fixes the electron scale and two different heavier-lepton graph polynomials simultaneously; it is not a blind prediction of G.

6. Charged-Lepton Shells as a Correlated Check

The electron is the $N = 0$ charged one-bit ground defect. The generation test deletes N labels and studies the closure response of the remaining alphabet. For $p_{N,\eta} \propto e^{-\eta K_N^2}$,

$$I_N(\eta) = \mathrm{Var}_{N,\eta}(K_N^2) = \langle s_N | s_N \rangle, \quad |s_N\rangle = \sum_b \sqrt{p_{N,\eta}(b)}(K_N^2(b) - \langle K_N^2 \rangle)|b\rangle.$$

Every four-label alphabet has $I_3 = 0$: every injective assignment uses all four labels and K^2 is permutation-invariant. Five- and six-label alphabets retain positive dispersion. Identifying one lepton grade with each nonzero closure-response grade yields $N = 0, 1, 2$. Excluding species built from other operators remains part of this particle identification.

The mass register uses a separate specified operation: each shell copies a six-label orthonormal encoding with injective four-port assignments and two orientations,

$$\Omega_1 = 2P(6, 4) = 720.$$

A rank-one singlet projector alone does not select that encoding: for $|u\rangle = 7^{-1/2} \sum_m |m\rangle$, all seven vectors $(I - |u\rangle\langle u|)|m\rangle$ remain nonzero. The model chooses a six-dimensional register isometry into $u^\perp$ and uses the same encoding at every shell.

Under memoryless renewal, the conditional mean return time to state b is $1/p_b$. A stationary choice $B \sim p$ therefore gives the exact arithmetic recurrence identity [3]

$$\mathbb{E}_B \mathbb{E}[T_B \mid B] = \sum_b p_b/p_b = 720.$$

For the full-support N-shell register it gives 720^N, even for correlated stationary weights. The geometric mean of conditional mean returns is instead e^{H_1}, with

$$\ln 720 - H_1 \simeq 0.00508\text{–}0.00782.$$

Thus 720 is an arithmetic support statistic, not an approximation to the admissibility Shannon support. Its identification with an inertial mass factor is a matching premise.

For coherent marks $|J_N\rangle = \sum_{r=1}^N |a_r\rangle$, the singlet norm is $\langle J_N|u\rangle\langle u|J_N\rangle = N^2/7$. The selected orientation-summed overlap is $q_s = 2/7$. One common centered Gaussian scalar fluctuation X_s, with variance $v_s = -2 \ln q_s$, realizes the positive overlap

$$\mathbb{E}[e^{iNX_s}] = e^{-N^2 v_s/2} = q_s^{N^2}.$$

Independent shell fluctuations would instead give q_s^N. On grades $0, 1, 2$, the prescribed log-overlap mass operator is

$$\boxed{\widehat{M} = m_e \exp[\widehat{N} \ln 720 + \widehat{N}^2 \ln(2/7) + \ln Z_{\widehat{N}}]},$$

where $Z_0 = 1$, $Z_1 = 1 + \zeta_*$ and $Z_2 = (1 + \zeta_*)(1 + 2\zeta_*/7)$. Its eigenvalues are

$$\frac{m_\mu}{m_e} = 720\frac{2}{7}Z_\mu = 206.768280237, \qquad (-0.535\sigma),$$

$$\frac{m_\tau}{m_e} = 720^2\left(\frac{2}{7}\right)^4 Z_\mu Z_{\tau,2} = 3477.343310, \qquad (-0.125\sigma).$$

The tau comparison is 1776.9188 MeV against the Particle Data Group (PDG) value 1776.93 ± 0.09 MeV [2]. Within the specified menu of 43 rationals in $[0, 1]$ with denominator at most eleven, only $2/7$ survives the one-standard-deviation tau window; $3/11$ gives -1.436σ. This finite menu has no probability measure over microscopic models.

Section 8 embeds $\widehat{M}$ in a selected covariant matter action. Section 14.6 supplies a finite orbital kinetic operator and native clock for one marked gap, with exact refinement on its encoded subspace. Extending that construction to the shell multiplets, physical fermion pole masses, and radiatively stable ratios remains open. Composite hadrons require a dressed, vacuum-subtracted quantum-chromodynamic (QCD) entanglement budget, $m_H = \kappa_m(\ell_H)S_{\mathrm{ent,H}}^{\mathrm{dressed}}$; neutrino and quark assignments remain open.

7. Edge Stiffness and the Exact Source Map

The tetrahedral identity

$$\sum_{i=1}^{4} \hat{n}_i \hat{n}_i^{\mathsf{T}} = \frac{4}{3} I_3$$

gives the transverse fraction $2/3$ and hence

$$\boxed{J_{\mathrm{bare}} = \frac{2}{3}\eta_*}, \qquad \boxed{J_{\mathrm{eff}}^{\mathrm{tree}} = \frac{J_{\mathrm{bare}}}{3} = \frac{2\eta_*}{9}}.$$

A minimal finite-return completion takes seven sector-diagonal returns plus one projected singlet,

$$\Sigma_{\mathrm{ret}} = 7 + \frac{2}{9} = \frac{65}{9},$$

so

$$J_{\mathrm{eff}}^{\mathrm{ren}} = \frac{J_{\mathrm{eff}}^{\mathrm{tree}}}{1 + J_{\mathrm{eff}}^{\mathrm{tree}}\Sigma_{\mathrm{ret}}} \simeq 0.00633348.$$

This loop operator is conditional: its algebraic resummation is fixed once the return operator is chosen, but an independent microscopic graph derivation of the relative diagonal/singlet weights remains an audit task. It cancels from the source-to-stiffness ratio below.

Reading the lattice weight as a Euclidean action and choosing the static graph convention $\Delta V_4 = L_*^4/c$ (time in seconds) gives the continuum occupancy stiffness

$$\gamma_Q = \frac{4\hbar c}{3L_*^2} J_{\mathrm{eff}}^{\mathrm{ren}}.$$

With horizon field normalization $S = \pi Q_{\mathrm{occ}}$,

$$\boxed{\gamma = \frac{4\hbar c}{3\pi^2 L_*^2} J_{\mathrm{eff}}^{\mathrm{ren}}}.$$

The source side closes independently. Excluding one of seven admissible labels removes exactly one-seventh of the isotropically averaged local partition weight, giving

$$\Delta S_{\mathrm{def}}^{\mathrm{ann}} = \ln\frac{7}{6}.$$

This is the annealed benchmark. A fixed excluded face-label type costs $-\ln P(A_{\ell f})$; averaging those costs differs from the benchmark by 5.7353×10^{-6} nats. The face marginal itself is not isotropic to that precision: $\max_m |p_m - 1/7| = 0.00414405$. The on-site Green function of the four-valent diamond adjacency is the exact lattice invariant

$$G_{\mathrm{tet}}(0) = \frac{3\Gamma(1/3)^6}{2^{14/3}\pi^4} = 0.4482203943883814\ldots.$$

The same tetrahedral frame gives a local transverse response $G_\perp = (2/3)G_{\mathrm{tet}}(0)$. Defining defect-entropy density by

$$\sigma_{\mathrm{def}} = \frac{\rho}{\kappa_m(L_*)},$$

the Green-matched continuum equation is

$$\boxed{\nabla^2 \delta S = -\frac{3L_*}{4G_{\mathrm{tet}}(0)}\sigma_{\mathrm{def}}}$$

and therefore

$$\boxed{\frac{\kappa}{\gamma} = \frac{3L_*}{4G_{\mathrm{tet}}(0)\kappa_m(L_*)}.}$$

At the cutoff,

$$\kappa_m(L_*) = \frac{\hbar}{cL_* \ln 2}.$$

The source theorem and bridge give, in fixed entropy units,

$$G = G_* \frac{S_0}{S_\infty}, \qquad S_0 := \frac{3\ln 2}{32\pi G_{\mathrm{tet}}(0)} = 0.0461482516\ldots.$$

The selected Einstein/horizon matching imposes $S_\infty^{\mathrm{cell}} = S_0$, hence $G = G_*$. A physical background βS_0 at fixed source coefficients gives G_*/β. Entropy-unit changes $S' = bS$, $\gamma' = \gamma/b^2$, $\kappa' = \kappa/b$ preserve $\kappa/(\gamma S_\infty)$. Neither this invariance nor the loop cancellation derives the background matching. The graph volume $\Delta V_4 = L_*^4/c$ and the angular model's G_0 require a common state, volume operator, and action normalization to determine an interacting G_R.

Source reduction must retain the complete generating functional. For the quadratic action $\frac{1}{2}x^T A x + x^T B y + \frac{1}{2}y^T D y - j_x^T x - j_y^T y$, $D > 0$, integrating out y gives

$$\begin{aligned} A_{\mathrm{eff}} &= A - BD^{-1}B^T, & j_{\mathrm{eff}} &= j_x - BD^{-1}j_y, \\ c_{\mathrm{eff}} &= -\tfrac{1}{2}j_y^T D^{-1} j_y + \tfrac{1}{2}\log\det D. \end{aligned}$$

Source derivatives include c_{eff} and every source-dependent coefficient. Sequential and joint elimination agree. Lean verifies the square-completion algebra; Gaussian integration uses the specified measure. The same requirement applies to source forces, clock response, and curvature normalization.

8. Einstein Gravity Is the Parent of the Closed Static Branch

The ordinary longitudinal branch has no independently varied scalar-gravity action. Its parent is Einstein–Hilbert plus minimally coupled matter,

$$I_0[g,\psi] = \frac{c^3}{16\pi G}\int d^4x\sqrt{-g}(R - 2\Lambda) + I_{\mathrm{GHY}} + I_{\mathrm{matter}}.$$

Here I_{GHY} is the Gibbons–Hawking–York boundary term. In static Newtonian gauge,

$$ds^2 = -\left(1 + \frac{2\Phi}{c^2}\right)(dx^0)^2 + \left(1 - \frac{2\Psi}{c^2}\right)\delta_{ij}dx^i dx^j,$$

expansion through quadratic order gives

$$I^{(2)}_{0,\mathrm{scal}} = \int dt\, d^3x \left[\frac{(\nabla\Psi)^2 - 2\nabla\Phi\cdot\nabla\Psi}{8\pi G} - \rho\Phi \right].$$

Varying the lapse and spatial scalar gives

$$\nabla^2\Psi = 4\pi G\rho, \qquad \nabla^2(\Phi - \Psi) = 0.$$

For an isolated solution with regular interior matching and no independent harmonic boundary data, asymptotic flatness gives

$$\boxed{\Phi = \Psi},$$

and eliminating Ψ gives the Newton action

$$\boxed{I_{\mathrm{Newton}}[\Phi] = \int dt\, d^3x \left[-\frac{(\nabla\Phi)^2}{8\pi G} - \rho\Phi \right].}$$

The static capacity coordinate is

$$I^{\mathrm{static}}_{\mathrm{cap}} = \int dt\, d^3x \left[-\frac{\gamma}{2}(\nabla\delta S)^2 + \kappa\rho\,\delta S \right].$$

The field redefinition

$$\boxed{\delta S = -\frac{2S_\infty}{c^2}\Phi}$$

turns this into a field-independent multiple of I_{Newton} provided

$$\boxed{G = \frac{c^2\kappa}{8\pi\gamma S_\infty}.}$$

Explicitly,

$$I^{\mathrm{static}}_{\mathrm{cap}} = Z_S I_{\mathrm{Newton}}, \qquad Z_S = \frac{2\kappa S_\infty}{c^2}.$$

Thus δS is a coordinate on the reduced Einstein constraint surface, not an extra radiative scalar.

Why the ordinary branch cannot be a second canonical scalar. The bridge is linear in the matter source: $\delta S = O(\rho)$ and $\Phi = O(\rho)$. By contrast, the canonical stress of an independent scalar begins quadratically,

$$T^{(\delta S)}_{\mu\nu} \sim \partial_\mu\delta S\,\partial_\nu\delta S = O(\rho^2),$$

so it cannot be the origin of the required $O(\rho)$ gravitational response. Nor is an explicit trace coupling $\delta S\, T^\mu{}_\mu$ acceptable as the ordinary source: it would fail to source classical radiation in the same universal way and would duplicate the metric response already present in the Einstein equation. The full stress tensor must therefore enter through the one physical metric.

Several simple auxiliary completions fail these requirements. An algebraic Arnowitt–Deser–Misner (ADM) multiplier $\lambda(N^2 - q)$ gives $\lambda = 0$ when q has no independent bulk dynamics, so it merely renames the lapse. Supplying q with genuine kinetic or higher-derivative dynamics introduces an additional scalar/preferred-foliation mode and a new stability audit. Scalar–tensor or disformal matter coupling likewise adds at least one scalar and generically introduces fifth-force, slip, and PPN obligations absent from the closed branch. Hence, under the stated requirements of universal stress-tensor coupling, no extra radiative degree of freedom, and the exact weak-field action identity, the metric-only/composite-capacity route is the minimal parent. This is a minimality result for the ordinary branch, not a proof that every substrate excitation is metric-only.

After imposing the physical canonical matching of Section 7,

$$\frac{\kappa}{\gamma} = \frac{3L_*}{4G_{\rm tet}(0)\kappa_m(L_*)}, \quad \kappa_m(L_*) = \frac{\hbar}{cL_* \ln 2}, \quad S_\infty^{\rm cell} = \frac{3\ln 2}{32\pi G_{\rm tet}(0)},$$

so the weak-field expression collapses identically to

$$\boxed{G = \frac{c^3 L_*^2}{\hbar} = G_*.}$$

This is one conditional normalization chain: the electron transfer and speed identification fix L_*, while the physical matching sets $G = G_*$. The source representation checks that same matching rather than determining it independently.

For a point source,

$$\nabla^2 \delta S = -\frac{\kappa}{\gamma}\rho \quad \Rightarrow \quad \delta S(r) = \frac{\kappa M}{4\pi\gamma r},$$

so the bridge gives

$$\boxed{g(r) = \frac{GM}{r^2}}.$$

Because the parent remains Einstein gravity, the baseline vacuum PPN values are

$$\boxed{\gamma_{\rm PPN} = \beta_{\rm PPN} = 1}$$

with the remaining standard PPN parameters zero under the usual assumptions.

Coupled matter and gravity on the selected branch. For the shell multiplet $\Psi = (\psi_0, \psi_1, \psi_2)^T$, a minimal leading completion is

$$I_{\rm matter} = \int d^4x\sqrt{-g}\,\bar{\Psi}(i\hbar c\,\gamma^a e_a{}^\mu D_\mu - c^2\widehat{M})\Psi, \qquad \{\gamma^\mu, \gamma^\nu\} = -2g^{\mu\nu} I$$

for the $(-,+,+,+)$ signature and displayed $i\gamma^\mu D_\mu$ convention. Each mass eigenstate then has $E_N^2 = c^2\mathbf{p}^2 + m_N^2 c^4$, while metric variation sources gravity with the same mass operator. This continuum action specifies the relativistic benchmark. The cochain operator in Section 14.6 realizes a finite positive-energy orbital sector and native mass clock. Coupling fermions through the frame oriented by the record order makes the refreshed lift bit inert. Chiral species, a universally selected common speed, and radiative stability remain open.

9. Carrier-Resolved Galactic Branch

The galactic excess is a coarse response of retained source records. Let e_P be sharp selectors in the retained label algebra, with $e_P e_Q = \delta_{PQ} e_P$ and $\sum_P e_P = 1$. For support- and source-integral-preserving conditional expectations E_P,

$$\Pi_P X = E_P(e_P X), \qquad \Pi_P^2 = \Pi_P, \qquad \Pi_P \Pi_Q = 0 \ (P \neq Q).$$

Active labels form an antichain. Nested expectations obey $E_P E_A = E_P$, and descendant selectors sum to e_P. A galaxy and its stars therefore receive only one transverse allocation; the full stress enters Einstein gravity once.

Spatial reconstruction C is a separate positive, mass-preserving map. Its profiles may overlap; smooth $\chi_P \geq 0$, $\sum_P \chi_P = 1$, weight the constitutive energies. They are not label projectors. Varying the reconstructed source gives the volume-weighted adjoint force $C^\dagger$; moving profiles also contribute derivatives of C and χ_P. The microscopic carrier selection, thermal Hamiltonian, and metric contact remain matching premises.

9.1 Acceleration scale and RAR

The response cell is the tilt mode of the transverse frame. The projected baryonic field is its restoring stiffness and its inertia lever is $\ell = g_{\rm share,eff}\, c/H$, the apparent-horizon radius multiplied by the sharing entropy of the cell, so $\omega = \sqrt{|\nabla\Phi_b|/\ell}$. Thermalized in the apparent-horizon bath at $k_B T_H = \hbar H/2\pi$, the mode has the gap $x = \hbar\omega/k_B T_H = 2\pi\omega/H = \sqrt{|\nabla\Phi_b|/a_0}$ with

$$\boxed{\epsilon := \frac{g_{\rm share,eff}}{4\pi^2} = 0.187945\ldots, \qquad a_0 = \epsilon c H_0.}$$

The standard linear-depletion fit gives $H_0 = 68.458\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $a_0 = 1.25005 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. The exponential-allocation fit with massless neutrinos, linear lensing and a low-redshift supernova offset gives $H_0 = 68.93885586416\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $a_0 = 1.25883 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. The Planck reference value gives $1.23073 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. For a lever in apparent-horizon radii the factor $4\pi^2$ is the square of the 2π in T_H. The tilt-mode identification, its horizon thermalization, and the lever are the matching inputs of the radial-acceleration relation. The lever coefficient is the ratio of the register enthalpy, $g_{\rm share,eff} k_B T$ at zero chemical potential, to the oriented record of entropy s that a static frame holds at the Unruh temperature $k_B T_U = \hbar g \cos\delta/2\pi c$ of the projected baryonic field, whose free energy $-s\, k_B T_U$ pulls the frame toward the field. The closure condition $\eta_* \langle K^2\rangle = 3/2$ gives the transverse pair that the tilt rotates exactly one nat, so the coefficient is $g_{\rm share,eff}$, and the RAR scale requires $0.85 < s < 1.39$. The Unruh reading of the baryonic part of the field is the premise. The mode is a collective excitation of the frame and carries no committed defect, so ordinary matter keeps equal inertial and gravitational mass and test-body free fall is unchanged. The occupation is a property of the equilibrium state, reached through the stock contacts, and requires no completed oscillation of a mode whose period near a_0 is about $4\pi^2/H$; the drift of H during that period would appear as a lag of $a_0(z)$ behind $H(z)$. Milgrom [25] related the MOND acceleration scale to the de Sitter horizon through the Unruh temperature of accelerated observers, with a different interpolation; the tilt mode gives the exponential interpolation and the coefficient $g_{\rm share,eff}/4\pi^2$.

For one active carrier, Φ_b denotes the projected baryonic potential satisfying $\nabla^2\Phi_b = 4\pi G C \Pi_P \rho_b$. The total potential is the full Einstein potential $\Phi_N[\rho_b]$ plus the carrier excesses $\sum_P \phi_{\perp,P}$. Define

$$x = \sqrt{\frac{|\nabla\Phi_b|}{a_0}}.$$

The cell Hamiltonian $H_{\rm cell} = \hbar\omega : N := k_B T_H x : N :$ has free energy

$$F_{\rm th} = k_B T_H \ln(1 - e^{-x}), \qquad \frac{\partial F_{\rm th}}{\partial E} = n_B(x) = \frac{1}{e^x - 1}.$$

The deep regime is the equipartition limit $n_B \simeq 1/x$ of the mode, and the Newtonian regime is its Wien suppression. To make the response conservative for nonspherical systems, set

$$z = \frac{|\nabla\Phi_b|^2}{a_0^2}, \qquad \mathcal{Q}_\perp(z) = \int_0^z \frac{ds}{e^{s^{1/4}} - 1},$$

and use the quasilinear modified Newtonian dynamics (QUMOND) auxiliary action. Variation gives

$$\boxed{\nabla^2\phi_\perp = \nabla\cdot[n_B(x)\nabla\Phi_b]\,.}$$

This slowly varying source-direction truncation fails across a thin-disk midplane. A nematic director $\hat{\mathbf{e}} \equiv -\hat{\mathbf{e}}$ gives $\mathbf{J}_\perp = n_B(\widetilde{x})(\hat{\mathbf{e}}\cdot\nabla\Phi_b)\hat{\mathbf{e}}$, with $\widetilde{x} = (|\hat{\mathbf{e}}\cdot\nabla\Phi_b|/a_0)^{1/2}$. The selected stiffness is $Z_\theta = 2c_\theta\Phi_R^2$, where Φ_R is the carrier's stock potential; the tested baryonic proxy uses $2\Phi_b^2$. Spherical exactness fixes the saddle-splay coefficient to unity:

$$\mathcal{S}[\hat{\mathbf{e}}] = |\partial\hat{\mathbf{e}}|^2 + \partial_i\hat{e}^a\partial_a\hat{e}^i - (\nabla\cdot\hat{\mathbf{e}})^2 = (\lambda_1 - \lambda_2)^2 + |(\hat{\mathbf{e}}\cdot\nabla)\hat{\mathbf{e}}|^2.$$

Here $\lambda_{1,2}$ are the principal splays. For a radial director, $\mathcal{S} = 0$ and the transverse elastic variation vanishes even with angularly varying stiffness. Nonspherical thermal forcing still requires a director solution. The baryonic-proxy auxiliary equation is $\nabla^2\phi_\perp = \nabla\cdot\mathbf{J}_\perp + \Phi_b\mathcal{S}$; varying the stock stiffness instead gives the mechanical contribution $\nabla^2 U_\theta = c_\theta\Phi_R\mathcal{S}$. A common stock/polarization dynamics and embedded-carrier potential reference remain completion data.

For the unprojected spherical target,

$$\boxed{g_{\rm obs} = \frac{g_{\rm bar}}{1-\exp[-\sqrt{g_{\rm bar}/a_0}]}}$$

with

$$g_{\rm obs}\to g_{\rm bar}\quad (g_{\rm bar}\gg a_0),\qquad g_{\rm obs}\to\sqrt{a_0 g_{\rm bar}}\quad (g_{\rm bar}\ll a_0),$$

and therefore $v^4\to GM_b a_0$.

9.2 Metric, lensing, and local limit

The leading quasistatic no-slip contact is

$$\boxed{h^\perp_{\mu\nu} = -\frac{2\phi_\perp}{c^2}(\bar g_{\mu\nu}+2u_\mu u_\nu),}$$

so $\Delta\Phi_\perp = \Delta\Psi_\perp = \phi_\perp$. Dynamics and lensing infer the same excess. In the local source-direction truncation,

$$\rho_{\rm lens} = \rho_b + \frac{1}{4\pi G}\nabla\cdot[n_B(x)\nabla\Phi_b].$$

For a deep-regime point mass, $g = \sqrt{GMa_0}/r$ and

$$\alpha_\infty = \frac{2\pi\sqrt{GMa_0}}{c^2}.$$

Carrier nesting makes the Milky Way the active transverse carrier in the Solar System, while the Sun and planets remain resolved Einstein descendants. Hence at leading quasistatic order

$$\boxed{Q_2^{\odot,\perp}=0},$$

and wide binaries remain Newtonian apart from the smooth Galactic tide. Nonlinear covariance of the record projector remains open.

Positive stock and retained angular information. For stock mass $M_R>0$ and a finite-budget target $\varrho_{\rm tar}$ of that total mass, the admissible target is

$$\rho_R^{\rm adm} = \frac{M_R[\varrho_{\rm tar}]_+}{\int[\varrho_{\rm tar}]_+ d^3x}.$$

It preserves mass and positivity. For nonincreasing spherical baryonic density and $M_R\ge M_b$, its total-force correction is at most 1.5311×10^{-4}. Positive spherical targets admit circular-shell kinetic equilibria; collective stability and formation require further analysis. Kinetic stock and director polarization must share one force law without adding a second full halo.

A finite radial update $w_{ia}^+ = (1-s)w_{ia} + sp_i m_a$, with $m_a=\sum_i w_{ia}$, $p_i\ge0$, $\sum_i p_i=1$, and $0\le s\le1$, preserves positivity, mass, and angular-bin marginals while contracting radial deviations. Eight Lean theorems verify these identities. The physical target, rate, and energy transfer remain inputs. Such relaxation can retain orientation; gravitational torques still evolve it.

Stock selection. The response cell has gap Θx, with $\Theta = k_B T_H$, and mean occupation $n_* = 1/(e^x - 1)$. With the stock carrying the response, the occupation is the stock field in baryonic units, $n = g_R/g_b$, and the excess free energy of a cell is $\Theta D(n\|n_*)$, with

$$D(n\|n_*) = n \ln \frac{n}{n_*} - (1+n) \ln \frac{1+n}{1+n_*} \ \geq 0.$$

Equipartition with the excess field, $\mathcal{N}\Theta = g_b^2 n_*(1+n_*)/4\pi G$, makes the Gaussian limit of $\int \mathcal{N}\Theta D \, d^3x$ equal to the field mismatch $(8\pi G)^{-1} \int |\nabla\Phi_R - \mathbf{J}|^2$. Local detailed balance with the horizon bath suppresses an uphill orbit change by $\exp(-|\Delta F|/\Theta)$ with $|\Delta F|/\Theta \gg 1$, so momentum-conserving pair contacts proceed when they lower the orbit-averaged mismatch. The cell energy $\mathcal{N}\Theta x n_*$ pays their kinetic change; per unit supported stock it tends to V_f^2 in the deep regime. The budget-constrained minimiser is the exponential law inside the budget edge with a Keplerian exterior. Spherical runs from abundance-matched NFW stock lie 0.009–0.032 dex rms from the law over 5–200 kpc for Milky-Way models after 8–12 Gyr, against 0.10 dex without contacts, and 0.007–0.05 dex for dwarf and massive-spiral models. The identification $n = g_R/g_b$, equipartition, a pair-kinetic contact rate proportional to local stock density and pair speed, and the record identification of Section 10.2, which coincides with the retained carriers of the release sector, are premises; a microscopic exchange operator and cosmological assembly remain open. Lean checks the relative-entropy positivity, equipartition matching, zero-temperature limit, finite H-theorem, pair-impulse conservation and constrained minimiser.

Resolved-source Milky Way test. The local contact gives 1792.2 $\mathrm{km^2\,s^{-2}}$ at $(8.17, 1.055)$ kpc for the flat-inner-disk model of Wang et al. [4], against 1294.8 ± 54.0. With an RAR-consistent disk the director gives 1401 or 1302, depending on stellar surface density; its 43-point $\chi^2 = 135.7$ or 139.5 matches NFW controls near 135. These are development postdictions. Old-disk tests in NGC 6946 and NGC 628 [152, 153] give rotation/dispersion mass agreement at 1.0σ and 0.3σ, versus 2.6σ and 2.9σ for QUMOND; a maximal disk with a round halo also passes.

The shape compilation favors near-roundness after allowing inter-study scatter. The aligned potentials erase stellar-debris tilt [154] in 0.15–0.25 Gyr. Three conditional live-stock realizations with 131072 samples, self-gravity and reciprocal torque on a rigid reacting disk retain stock tilts 39.6–42.4° and stellar tilts 21.4–22.3° over 10–40 kpc, averaged over 7.5–8 Gyr. The matched spherical 6–60 kpc cut gives 22.6–24.7° for stars. Initial 43° stock tilt, quadrupole and orbit population specify the candidate; a_0 is unchanged. The 8% precession difference is relative to its construction rate. Relative-axis drift, force convergence, disk bending, coupled director response, and formation remain open.

10. Causal Transport and Clusters

10.1 Transport

The telegrapher completion preserves the static limit:

$$\tau_0 \ddot{\phi}_\perp + \dot{\phi}_\perp - D\nabla^2 \phi_\perp = -4\pi G D \rho_\perp, \qquad D/\tau_0 = c^2, \quad \tau_0^{-1} = H_0.$$

Its Fourier response and poles are

$$F(\omega, k) = \frac{Dk^2}{Dk^2 - \tau_0 \omega^2 - i\omega}, \qquad s_\pm = \frac{-1 \pm \sqrt{1 - 4\tau_0 D k^2}}{2\tau_0}.$$

For $k > 0$, $F(0,k) = 1$ and both poles decay; $k = 0$ includes a neutral mode. The characteristic speed is c. At $\omega \sim vk$, inertial and damping corrections are $O(v^2/c^2)$ and $O[v/(Dk)]$. Underdamped transients decay on $2H_0^{-1}$, so the quasistatic branch requires explicit initial data.

At a common temperature, symmetrized noise and dissipative response obey

$$N(\omega) = \hbar \coth\left(\frac{\hbar\omega}{2k_B T}\right) \mathcal{D}(\omega), \qquad \mathcal{D} = -\operatorname{Im} \chi^R.$$

The classical factor $2k_BT/\omega$ requires $\hbar\omega \ll k_BT$, hence $\omega \ll H/(2\pi)$ at horizon temperature. A finite-bandwidth causal response and its quantum noise must be paired; coupling and independent-cell count remain unspecified. Reconstruction $\chi_c = C\chi_f C^\dagger$, $N_c = CN_fC^\dagger$ preserves this relation and positivity. For a fine generator L, autonomous Markov closure requires $CL = L_cC$; unresolved modes generally supply memory.

10.2 Cluster stock and carrier support

Baryons and a positive retained stock move in one potential, $\nabla^2\Phi = 4\pi G(\rho_* + \rho_g + \rho_C)$, so lensing and dynamics measure the same inventory. A fully released closed cohort has

$$\boxed{k_C = 5.333\, e^{-\eta_*} = 5.1760752, \qquad M_{\rm grav} = (1 + k_C)B = 6.18\, B}$$

for coextensive retention, in the formation mass convention of the cosmological fit; the bare convention gives $e^{-\eta_*}/\epsilon = 5.164$. Inside an aperture the retained fractions of baryons and stock can differ, $C/B = k_C f_C^{\rm ret}/f_b^{\rm ret}$, and their histories require forward evolution.

The stock selection of Section 9 acts through the cells of active carriers, on the stock each carrier carries. A carrier's records are identified with its condensed baryons, stars and cold gas. Shock-heated gas whose cooling time exceeds its dynamical time relaxes to the Gibbs state of its potential and holds no independent record. A carrier's support ends where its records stop dominating the baryonic field. For an isolated carrier this is where they stop dominating the enclosed baryons. A satellite whose host's baryonic field exceeds its own at its half-light radius has no support of its own and keeps its pre-infall stock; for the eight classical Milky-Way dwarfs this assignment gives $\chi^2 = 10.4$ on their GravSphere central densities, against 17.2 for retained and 44 for relaxed stock throughout. Dwarfs whose progenitors never crossed the release gate keep their committed, CDM-like stock; over 22 Local Group dwarfs this combined assignment gives $\chi^2 = 21.7$ (25.4 at a $14\,{\rm km\,s^{-1}}$ gate) for half-light masses, against 24.0 for ΛCDM halos with supernova heating, and the gate value is left open by these data. Its cells act on the stock bound to the potential of its records and the stock it carries. Around an isolated lens the support ends where hot gas outweighs the condensed baryons, and the uncarried stock outside it keeps the halo profile; on the KiDS-1000 isolated-lens bins this matches truncated NFW halos within the halo-model nuisances ($\chi^2 = 140$ against 133 over 60 points). In the seven X-COP clusters with stellar profiles, condensed baryons are 2.5–5% of the baryons inside R_{500}, and the central galaxy's support ends at 0.04–$0.07\,R_{500}$. Cluster stock outside that support enters no cell, evolves as collisionless matter, and keeps $1 + k_C = 6.18$, between the X-COP medians 6.10 at R_{200} and 6.45 at R_{500} [155].

A response acting on cluster stock fails in either direction. At R_{500}, where $g_{\rm bar} = 0.05$–$0.07\,a_0$, imposing the law on the existing stock removes 20–35% of it, and adding the law to the cohort stock gives 9.4–10.2. Galaxy outskirts that obey the law lie at the same $g_{\rm bar}/a_0$, so the switch is not a function of acceleration. Contact runs on the seven clusters, started from the observed profiles, keep the mass inside R_{500} within 0.1% and inside $0.1R_{500}$ within 1.5% of the no-contact control over 10 Gyr; the same contacts with cells throughout the cluster leave 55–79% inside R_{500}. The central galaxy's carried stock relaxes toward its galactic-law value while its support holds seven to eleven times that value in uncarried stock, consistent with the observed departure of brightest cluster galaxies and group centrals from the rotating-galaxy relation. The X-ray ellipticals on the radial acceleration relation [156] are star-dominated over the fitted radii; in NGC 720 hot gas dominates the baryons only beyond about 200 kpc [157].

Gas shocks while stars and stock continue through an encounter, so gas and lensing peaks can separate. In the CANUCS Bullet reconstruction the subcluster halo centre lies 3.9 kpc from its brightest galaxy and the gas peak about 170 kpc away [158]; controlled encounters at 24–30 kpc softening cannot yet test that alignment. Resolved member galaxies, cosmological assembly, merger morphology, and the baryon–stock segregation $\eta_{\rm ret} = 0.52$–0.70 required by the A1795 and A2029 lensing targets remain open.

11. Cosmology: Homogeneous Mode, Saturated Capacity, and Release

11.1 Homogeneous mode

Write

$$S(x,t) = \bar{S}(t) + s(x,t), \qquad S_\infty(t) = \pi \frac{R_A(t)^2}{L_*^2}.$$

The local weak-field branch depends on fractional fluctuations and $\kappa/(\gamma S_\infty)$, while $\bar{S}(t)$ can affect the background on horizon scales. The proposed Hubble-tension mechanism is that a homogeneous trace-sensitive capacity mode becomes active near matter–radiation equality and reduces the sound horizon. The sign and timing are structurally plausible, but the joint Einstein–Boltzmann evolution has not been completed; this sector remains open/conditional. A fit of a proxy early-energy potential together with the release history, with the onset fixed at equality, $\omega_c = 5.333\,\omega_b$ and one added amplitude, gives an early-energy fraction 0.018 ± 0.010, $H_0 = 69.4 \pm 0.6\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $r_{\mathrm{drag}} = 146.4\,\mathrm{Mpc}$, with χ^2 below ΛCDM by 1.0 at the same parameter count; the tie excludes fractions of 0.05 and above ($\Delta\chi^2 \geq 13.5$) and hence $H_0 \simeq 73$.

11.2 Saturated committed-capacity phase

The adopted source-allocation cap is $1 + 1/\epsilon \simeq 6.32$ including the ordinary unit response. It cannot cap the galactic force ratio: the exponential RAR exceeds it whenever

$$y := g_{\mathrm{bar}}/a_0 < [\ln(1+\epsilon)]^2 = 0.0296616\ldots.$$

The recombination-era estimate $y \lesssim 6\times10^{-5}$ gives a formal unsaturated demand above 10^2. Mapping that demand to a finite committed allocation motivates saturation but is an additional cross-phase recruitment premise. For each Lagrangian baryonic source A, define

$$M_{c,A}^{\max} = \frac{M_A}{\epsilon}, \qquad \sigma_A = \epsilon \frac{M_{c,A}}{M_A} \in [0,1].$$

The pinned branch assumes every recruited source reaches $\sigma_A = 1$ while its coherent renewal phase remains single-stream. In this cosmological subsection use $c = 1$. Let $\phi = \tau_* \theta$ advance once per calibrated tick. Coherence gives $u_\mu = -\nabla_\mu \phi$, and one tick per proper-time tick forces

$$1 = \frac{d\phi}{d\tau} = -u^\mu u_\mu \quad \Rightarrow \quad \boxed{X = -\frac{1}{2}\nabla_\mu\phi\nabla^\mu\phi = \frac{1}{2}}.$$

Thus the leading constrained action

$$S_{\mathrm{comm}} = \int d^4x\sqrt{-g}\,\lambda(X - 1/2)$$

gives a pressureless, zero-sound-speed stress and $\rho_c \propto a^{-3}$ within the pinned coherent branch. The constraint is derived from saturation plus coherent tick normalization; the transition into that branch is not.

Full saturation gives the bare pre-release abundance

$$\boxed{\frac{\Omega_c^{\mathrm{wb}}}{\Omega_b} = \frac{1}{\epsilon} = 5.320684854, \qquad a_0 \frac{\Omega_c^{\mathrm{wb}}}{\Omega_b} = cH_0.}$$

At partial formation saturation the ratio is $\bar{\sigma}/\epsilon$. The specified formation map allocates capacity only to unlabelled conserved baryonic charge and debits the plasma by ρ_b/ϵ. Its proposed interval is between sphaleron freeze-out and the QCD crossover, before the relevant modes enter the horizon. The free-nucleon and helium conversion gives the formation input 5.333; the reference release history

lowers the present dust-to-baryon ratio to about 5.2284 for the reference exponential allocation. The bare identity above must be converted before comparison with that ratio.

The label ledger preserves or merges ancestry and removes cap together with released allocation. Once every conserved source is labelled, the specified transfer cannot recreate unlabelled charge or recruit primordial stock at any later temperature. This no-late-recruitment result assumes the ledger rule. A late allocation retains a positive clock deficit; relative motion can only increase it. The formation map is isometric; its microscopic vertex and the independent derivation of its rate remain open.

11.3 Collective carrier loss and dark-energy release

Shell crossing alone cannot release identity-free committed capacity: that register does not encode which identical stream owns it. The selected trigger is irreversible loss of the baryonic progenitor record. For each elementary record block,

$$\mathsf{E}_A = \mathbf{1}[\text{the baryonic carrier of } A \text{ loses its record}].$$

A partial-memory family $K_a(b,b') = a\delta_{bb'} + (1-a)p_*(b')$ retains information for every $a > 0$. The entropy maximum selects $a = 0$. The normalized nine-output isometry then gives unit conditional register transfer, $f_{\text{reroute}} = 1$, for a failed block. Identifying that register with the full allocated energy is a separate ledger premise.

The physical energy operators must include their conversion factors:

$$H_{\text{comm}} = \varepsilon_K \widehat{K}^2, \qquad H_{\text{rec}} = \frac{\varepsilon_\xi}{2}\sum_{a=1}^{3}\widehat{\xi}_a^2.$$

With the fresh normalized Gaussian record, $\langle \sum_a \xi_a^2/2\rangle = 3/2$. Refresh-first independence gives

$$\boxed{\eta_{\text{drain}} = \frac{\varepsilon_\xi}{\varepsilon_K}\frac{3/2}{\langle K^2\rangle_{\eta_*}} = \frac{\varepsilon_\xi}{\varepsilon_K}\eta_*}.$$

The selected common conversion $\varepsilon_\xi = \varepsilon_K$ yields $\eta_{\text{drain}} = \eta_*$. Neither equality of energy units nor the refresh-first ordering follows from Hilbert-space dimension alone.

Information stock and the cumulative kernel. For progenitor label A and retained matter observable Y_τ, define

$$K_{\text{mix}}(\tau) = 1 - \frac{I(A;Y_\tau)}{H(A)}.$$

Continuity and additive pricing assign K_{mix} to the reallocated fraction of surviving stock. Source-record identification and pricing are physical premises; the drain below determines the surviving energy. For a closed Lagrangian cohort under common advection and label-independent diffusion, let ρ be the total density, c_A the local mass fraction carrying label A, M_{gas} the cohort mass, and $\mathbf{D} \succeq 0$ the common diffusion tensor. The fully resolved position information obeys

$$\dot{I}(A;X) = -\frac{1}{M_{\text{gas}}}\sum_A \int \frac{\rho}{c_A}(\nabla c_A)^T \mathbf{D}\nabla c_A \, dV \le 0, \qquad K^X_{\text{mix}} = 1 - \frac{I(A;X)}{H(A)}.$$

Hence $0 \le K^X_{\text{mix}} \le 1$ and $\dot{K}^X_{\text{mix}} \ge 0$. This does not prove monotonicity for the coarse Y_τ in the physical stock law: even fixed two-site bins on a four-site cycle with labels $(0,0,1,1)$ give losses $0,1,0$ under successive reversible translations. The actual observable, changing cohort, boundary flux, and selective escape must therefore be checked in the cosmological zoom.

Let $F_Z(t_f)$ be the mass-weighted cumulative fraction entering an eligible retained well for the first time at t_f. The delayed release history is

$$\boxed{F_{\rm rel}(t) = \int^t K_{\rm mix}(t - t_f; M, z)\, dF_Z(t_f)}.$$

This is a joint mass-weighted first-entry integral when mass and entry redshift vary. A first self-gravitating progenitor-tagged run gives entry loss $K_0 \simeq 0.5$ and a 2–6 dynamical-time tail. Its linear-benchmark integral is $I_{\rm rel} = 76$ for three hosts at a 17 $\mathrm{km\,s^{-1}}$ gate and 93–97 for ten hosts at 14 $\mathrm{km\,s^{-1}}$, near the earlier PPF target 94.2116. Choosing the gate to match that target supplies no independent normalization. Gas convergence, labels, and retention gates remain physical inputs.

Ensemble clock and positive allocation. The homogeneous law is $Q(\phi) = \dot\rho_\Lambda = \eta_* \mathbb{E}_{\rm bg}[q](\phi)$ in a statistically homogeneous history state. A large-volume slice average estimates this ensemble quantity. Realized local release changes local records without instantaneously updating the prescribed vacuum at distant events. With $c = 1$, the local clock action is [85, 5]

$$S_{C\Lambda} = \int \sqrt{-g}\,[\lambda(X - \tfrac{1}{2}) - V_F(\phi)]\, d^4x, \qquad X = -\tfrac{1}{2}\nabla_\mu\phi\nabla^\mu\phi.$$

It gives $X = 1/2$, $u_\mu = -\nabla_\mu\phi$, $C = \lambda$, and $T^{\mu\nu} = Cu^\mu u^\nu - V_F g^{\mu\nu}$. The unit-gradient flow is geodesic, with $\nabla_\mu T_C^{\mu\nu} = \nabla^\nu V_F$. This is an irrotational single-flow completion; multistream stress needs further specification.

Write $C = \rho_u + R$, with local transfer $q = \Gamma_{\rm rel}\rho_u \geq 0$. Requiring the drain to preserve internal composition fixes $d_u = \rho_u Q/C$, $d_R = RQ/C$ uniquely for $C > 0$. Thus

$$\dot\rho_u + \theta\rho_u = -q - \rho_u Q/C, \quad \dot R + \theta R = q - RQ/C, \quad \dot F = \Gamma_{\rm rel}(1 - F), \quad F = R/C.$$

Both components remain nonnegative until total-stock exhaustion. The ensemble background has $F = F_{\rm rel}$ and, with $B = \rho_{c,0}^{\rm wb}$,

$$\boxed{a^3\bar C = Be^{-\eta_* F}, \qquad V_F(\phi) = \rho_{\Lambda,i} + \eta_* B \int_{t_i}^{\phi} a(t')^{-3} e^{-\eta_* F(t')}\, dF(t').}$$

The ensemble solution specifies this potential before variation. The initial condition $\rho_{\Lambda,i} = 0$ is adopted independently of vacuum centering. Intrinsically $p_\Lambda = -\rho_\Lambda$; a separately conserved reconstruction gives $w_{\rm eff} = -1 - \dot V_F/(3HV_F) \leq -1$ during accumulation.

Timing and the local domain. In the reference exponential history, by $z = 2$ cumulative comoving release is 52.17% complete and vacuum accumulation is 96.82% complete, owing to the a^{-3} weight. The drain after $z = 1$ is 0.4938% of initial stock. Local positivity requires

$$JC = J_i C_i - \int JQ\, dt \geq 0, \qquad \dot J = \theta J.$$

Under common baryon–capacity flow, a baryon density floor above $1 - e^{-\eta_* F(1)} = 0.01961844$ of the mean throughout release suffices. The full paper's O.3 nonlinear spherical histories ending at $\Delta_C = 0.2, 0.1$ stay positive; its specified growing-mode evacuation exhausts today at $\Delta_b \simeq 0.006835$. That endpoint is history-dependent. A density-dependent local shutoff generally produces $h^\mu{}_\nu\nabla^\nu V \neq 0$ and requires momentum exchange. The uniform-clock branch has zero transverse force and ends at exhaustion.

Linear perturbations and the tested benchmark. The clock fixes $\delta\dot{\phi} = \psi_g$ and $\delta\rho_\Lambda = \dot{V}_F\delta\phi$; vacuum is homogeneous on capacity-clock slices. Subtracting conserved reference dust $\rho_{\rm ref} \propto a^{-3}$ on the same flow gives the exact effective fluid $\rho_D = C - \rho_{\rm ref} + V_F$, $p_D = -V_F$, with rest-frame $c_s^2 = 0$ where its enthalpy is positive.

Direct geodesic perturbations with matched lensing and $A_L = 1$ give $\Delta\chi^2 = +0.85$ against ΛCDM for Planck high-ℓ+DESI DR2+DES-Dovekie with $\sum m_\nu = 0.06\,\mathrm{eV}$, nonlinear lensing and a free Planck calibration with prior 1 ± 0.0025. This linear-depletion benchmark uses $a^3\bar{C} = B(1 - \eta_* F)$ and $I_{\rm rel} = \int a^{-3} dF$, with five versus six cosmological parameters ($\Delta\mathrm{AIC} = -1.15$). The configuration with massless neutrinos, fixed calibration, nonlinear lensing and a free supernova offset below $z = 0.1$ gives $\Delta\chi^2 = -0.39$.

The exponential-allocation fit with massless neutrinos, fixed calibration, linear lensing and the low-redshift offset gives $\chi^2 = 2227.8006205$ against the matched ΛCDM value 2228.3158253, hence $\Delta\chi^2 = -0.5152$. Its release fit has $H_0 = 68.93885586416\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. These fits use the release histories and likelihood settings given above. With the retention gate taken from the halo occupation of feedback-resolved simulations, the measured mixing kernels and the reionization redshift of the fitted optical depth, the release history has no free parameter. That frozen history gives $\Delta\chi^2 = +1.07$ against ΛCDM in the first configuration and -0.28 with massless neutrinos and the low-redshift offset, with four cosmological parameters against six, and $H_0 = 68.42\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. An independent abundance test measures the formation and mixing inputs before fitting the cosmology; joint CMB-lensing reconstruction and nonlinear evolution extend the likelihood.

12. Clock Capacity, Strong Fields, and the Boundary

The surviving fraction $q = S_{\rm ent}/S_\infty$ lies in $[0, 1]$. Its weak-field relation is $q = 1 + 2\Phi/c^2 + O(c^{-4})$. For a static clock, continuous multiplicative composition and weak-field matching select $q_\tau = N^2$, with the Killing time normalized at the reference boundary. Write

$$ds^2 = -N^2c^2dt^2 + h_{ij}dx^i dx^j, \qquad s_\tau = S_\infty(1 - N^2).$$

At $\Lambda = 0$, Einstein's static lapse equation is

$$D^2N = \frac{4\pi G}{c^4}N(\mathcal{E} + \mathcal{S}), \qquad \mathcal{E} = T_{\mu\nu}n^\mu n^\nu, \quad \mathcal{S} = h^{ij}T_{ij}.$$

Here n^μ is the unit static normal. Consequently

$$\boxed{D^2 s_\tau = -\frac{8\pi G S_\infty}{c^4}\left(1 - \frac{s_\tau}{S_\infty}\right)(\mathcal{E} + \mathcal{S}) - \frac{|Ds_\tau|^2}{2(S_\infty - s_\tau)}.}$$

Its weak, nonrelativistic limit is $\nabla^2 s_\tau = -8\pi G S_\infty \rho/c^2$. The spatial metric and matter equilibrium equations complete the system. The coefficient identities $\kappa_{\rm phys} = c^2/(2S_\infty)$ and $\gamma_{\rm phys} = c^4/(16\pi G S_\infty^2)$ express a supplied Einstein normalization.

Spherical symmetry supplies a separate radial invariant. For $ds^2 = h_{ab}dx^a dx^b + R^2 d\Omega^2$, angular reduction of Einstein–Hilbert plus its boundary action gives

$$I_{\rm sph} = \frac{c^3}{4G}\int d^2x\sqrt{-h}[R^{2(2)}R + 2(\nabla R)^2 + 2] + I^{(2)}_{\rm matter} + I_\partial.$$

Define

$$\boxed{q_R = (\nabla R)^2 = 1 - \frac{2GM_{\rm MS}}{c^2 R}.}$$

The vacuum equations conserve $M_{\rm MS}$. In a connected, normalized, static vacuum exterior,

$$q_R = q_\tau = N^2 = 1 - \frac{2GM}{c^2 r}.$$

Inside spherical matter of mass density ρ and radial pressure p_r,

$$\frac{d}{dr}\log\frac{q_\tau}{q_R}=\frac{8\pi Gr}{c^2q_R}\left(\rho+\frac{p_r}{c^2}\right).$$

Thus the locally flat interior of a supported shell can have $q_R=1$ and a constant $q_\tau<1$. The clock redshift includes boundary reference data. The surface $q_R=0$ is a marginal sphere; equating it with substrate saturation during collapse and excluding $q_R<0$ require a separate boundary theory.

For a dynamical ADM slicing, with $c=1$, $K_{ij}=-\frac{1}{2}\mathcal{L}_n h_{ij}$ and $q_N=N^2$,

$$\begin{aligned}D^2q_N=2q_N[K_{ij}K^{ij}+4\pi G(\mathcal{E}+\mathcal{S})-\Lambda]\\-2N(\partial_t-\mathcal{L}_\beta)K+\frac{|Dq_N|^2}{2q_N}.\end{aligned}$$

The lapse depends on slicing and boundary conditions. A general capacity observable must specify its relational region, quantum state and algebra, clock, and reference data, without additional gravitational Cauchy data. For smooth perturbations of a faithful reference, $D(\rho_\epsilon\|\rho_0)=O(\epsilon^2)$; a differentiable function of relative entropy alone cannot give a generic linear gravitational response. The finite hard-exclusion identities use a different state change and remain valid.

The exterior Hawking temperature and Bekenstein–Hawking law retain their Einstein form. For coherent maximal fusion of N primitive $j=3$ faces,

$$A_N=24\pi\gamma_{\rm BI}NL_*^2,\qquad S_{\rm cut}=N\ln 2.$$

Comparison with $A_N/(4L_P^2)$ gives

$$\frac{G}{G_*}=\frac{6\pi\gamma_{\rm BI}}{\ln 2}.$$

On the matched $G=G_*$ branch this sets $\gamma_{\rm BI}=\ln 2/(6\pi)$ and $a_{\rm ch}=4\ln 2\,L_*^2$. The independently calibrated angular construction in Section 14.2 instead fixes $A_{\rm ch}=4\ln 2\,\hbar G_0/c^3$ before any equality with G_*. Its cut and bulk coefficients must renormalize consistently in one physical state.

For a convex spatial horizon with a Euclidean three-space representation, Cauchy's formula gives $\overline{A}_{\rm proj}=A/4$ [6]. Writing the matched entropy as $\overline{A}_{\rm proj}/L_*^2$ restates its normalization. It does not derive that entropy density or extend the projection formula to every dynamical horizon. Boundary dynamics, reflectivity, saturation formation, and generic rotating or charged capacity observables remain open.

13. Many-Pasts and the Finite Quantum Sector

For a decoherent history h with class operator C_h,

$$\mathcal{D}(h,h')=\mathrm{Tr}(C_h\rho_0C_{h'}^\dagger),\qquad p(h,P)=\mathcal{D}(h,h),\qquad p(h|P)=\frac{\mathcal{D}(h,h)}{\sum_{h'\in\mathcal{H}_P}\mathcal{D}(h',h')}.$$

Conditioning is performed only after the record P exists; it changes no Hamiltonian or class operator and introduces no retrocausal force.

On the stable retained sector, a positive real quadratic Hamiltonian selects an initial polar complex structure J_0. Full symplectic evolution transports it as $J(t)=S(t)J_0S(t)^{-1}$ and preserves the corresponding Hermitian product. The exact transport retains squeezing relative to an instantaneous polarization and the induced Born–Huang term. Within the selected complex-linear control realization, moving frames and a non-common frequency generate $\mathfrak{su}(9)$; the fixed-one-mark response algebra gives $\mathbb{C}^9$ and pure-state space $\mathbb{CP}^8$. Access to controls on a common polar fiber is an operational premise; general symplectic frame motion can change that fiber. Independent retained factors compose by tensor product. Geometric gauge projection is the identity on the capacity factor, so its

bridges survive; adjacent retained marks generate $\mathfrak{su}(81)$. Their bounded finite-range bridge generator, of norm 41/5, gives a quasi-local infinite-volume algebra, Lieb–Robinson dynamics, and GNS representations on bounded-degree hosts. The physical state and continuum particle interpretation remain open. The physical canonical form is twice the Kähler form in action units. The internal spin-3/2 leakage coefficient 7/20 is exact, but its galactic response identification remains open.

For the probability rule, reversible internal equivalence makes a branch weight depend only on its squared norm, $w = f(\|\psi\|^2)$. Complete mutually exclusive physical records add under coarse graining, so

$$f(x+y) = f(x) + f(y), \qquad f(x) \geq 0, \qquad f(1) = 1.$$

Finite additivity plus positivity fixes $f(x) = x$ on the physical probability domain. Hence

$$\boxed{p(\Pi|\psi) = \|\Pi\psi\|^2 = \langle\psi|\Pi|\psi\rangle}.$$

Reversible system–record coupling then gives the usual Kraus operators and POVMs; tracing a remote complete instrument leaves the local marginal unchanged, giving no-signaling.

Many-Pasts retains one realized present and the ordinary quantum history measure. Renewal replaces the closure register with the fresh state ρ_* each tick, so within one tick it is an exact relative-entropy H-theorem on that register, and the export structure fixes its orientation. The persistent geometry and defect sector evolves unitarily and keeps its superpositions. A recorded subsystem relaxes when an irreducible record-writing coupling acts on it, and on a microcanonical shell its present entropy then increases. The macroscopic second law follows from record growth, the minimal-record root and conditional typicality. Export keeps the global dilation reversible. This finite-network arrow does not set a cosmological relaxation time.

Exact recovery requires $\dim \mathcal{H}_{\mathrm{hist}}(N) \geq \prod_n d_n$ for independent exported alternatives. Finite total present capacity D_{now}, including history, with $d_{\min} > 1$ and no external archive give $N_{\mathrm{past}} \leq \ln D_{\mathrm{now}} / \ln d_{\min}$ and a minimal-record root. Its gravitational entropy and the physical supply of fresh history capacity remain open. In the specified number-conserving dynamics, $[H, n_m] = 0$ makes the released zero-defect sector absorbing. For number-changing completions the thermal reentry bound is finite, so exact exclusion over an unbounded future needs an additional restriction; the conserved-source ledger still forbids regeneration of primordial stock.

Renewal and spatial coherence. Replacement acts on the fast capacity register, while retained position and the marked Q fiber carry coherent transport. A dilation multiplies an unmeasured spatial off-diagonal by the overlap of the discarded records. A fresh marginal alone does not control this overlap. For a specified SWAP-reset dilation, two branches triggering n, m resets give

$$\gamma_{nm} = \operatorname{Tr} \rho_*^{|n-m|+1}.$$

One additional reset gives 0.0006036303 for the native state. This is a counterexample for that dilation.

A compatible fixed-background realization uses

$$\mathcal{E} = \mathrm{id}_{X,Q} \otimes \mathcal{E}_{\mathrm{background}}.$$

It renews the background independently of unmeasured particle position; proper-time phase accumulation causes no additional erasure. Retained-density and disjoint-reset-order errors are below 6×10^{-17} in the finite checks. For a graph-dependent bond vacuum $U_g|\mathrm{ref}\rangle = |\Omega_g\rangle$, the controlled frame

$$D = \sum_g |g\rangle\langle g| \otimes U_g$$

uncomputes to a common renewal frame before retained evolution and preparation of the outgoing frame. The encoded transfer then intertwines the retained operator exactly. Directly exporting $|\Omega_g\rangle$ would reveal the graph. Matching exported records for geometries with different numbers of physical events, and full refoliation, remain additional requirements.

14. Microscopic Geometry, Dynamics, and the Matter Clock

14.1 Representation factors and geometric transport

Two structures called "closure" must be distinguished. Geometric Gauss closure is an exact redundancy constraint: on a physical gauge-invariant state,

$$\widehat{\mathbf{C}}_{\mathrm{G}}|\psi_{\mathrm{phys}}\rangle = 0 \quad \Longrightarrow \quad \widehat{\mathbf{C}}_{\mathrm{G}}^{2}|\psi_{\mathrm{phys}}\rangle = 0.$$

Capacity closure is instead a physical statistical weighting of routed states. Its certified spectrum has the strict floor

$$\boxed{K^2 \geq \frac{122}{3} > 0}$$

on every state of the nonempty 1680-state ensemble. Therefore the two operators cannot be the same:

$$\boxed{K^2_{\mathrm{admissibility}} \neq C^2_{\mathrm{Gauss}}}.$$

The distinction changes the ensemble. If the Gibbs factor $e^{-\eta K^2}$ were built from the geometric Gauss Casimir and one projected to physical states, that factor would become unity, erasing the nonuniform ensemble and with it the stationarity equation that fixes η_*. Thus any microscopic realization that preserves both ordinary geometric gauge invariance and the finite numerical spine must keep the capacity data physical but inert under the geometric gauge redundancy.

The selected minimal completion therefore factorizes one cell as

$$\boxed{\mathcal{H}_{\mathrm{cell}} = \mathcal{H}_{\mathrm{geom}} \otimes \mathcal{H}_{\mathrm{cap}} \otimes \mathbb{C}^2_{\mathrm{orient}}.}$$

The geometric gauge action is

$$U_{\mathrm{gauge}}(h) = U_{\mathrm{geom}}(h) \otimes I_{\mathrm{cap}} \otimes I_{\mathrm{orient}},$$

so every capacity observable obeys

$$\boxed{[O_{\mathrm{cap}}, U_{\mathrm{gauge}}(h)] = 0}.$$

Consequently the injective 1680 count, K^2 spectrum, Gibbs weights, and η_* survive geometric gauge projection unchanged.

Relational frame geometry must include genuine transport. For reference links $U_{cc'}$ and local frames R_c,

$$g_{cc'} = R_c U_{cc'} R_{c'}^{-1}, \qquad g_{cc'} \mapsto h_c g_{cc'} h_{c'}^{-1}.$$

Loop holonomy is $R_c U_{\mathrm{loop}} R_c^{-1}$ and may carry curvature; a product of endpoint frames alone would be pure gauge.

The two bookkeeping factors nevertheless describe the same relational orientation on the physical branch. An equivariant lock

$$L_3 : V_3^{\mathrm{cap}} \to V_3^{\mathrm{geom}}, \qquad L_3 U_{\mathrm{cap}}(g) = U_{\mathrm{geom}}(g) L_3,$$

identifies their selected V_3 representations. Because both are irreducible, Schur's lemma gives

$$\boxed{L_3 = e^{i\phi} I}$$

after choosing a common relational basis. The existence of the lock is the selected realization; once it exists, it introduces no continuous physical coupling.

On the selected maximal channel, multiplicity one gives a canonical, tree-independent blocking map

$$\boxed{F_N : V_3^{\otimes N} \to V_{3N}}, \qquad F_N|3,\mathbf{n}\rangle^{\otimes N} = |3N,\mathbf{n}\rangle.$$

Proper Engle–Pereira–Rovelli–Livine (EPRL) selection isolates the gravitational asymptotic sector. The selected metric-Regge limit has a reflection-positive Gaussian TT covariance and the bosonic Fock representation over $L^2(\mathbb{R}^3)\otimes\mathbb{C}^2$, with two massless helicity-two modes. This closes the free linearized branch; an interacting native phase and one temporal/spatial light cone remain tests. The factorized capacity decoration preserves this spectrum.

For $J^\dagger J = I$ and $H_f J = J H_c$, the encoding $\iota(A) = JAJ^\dagger$ preserves products, traces, and positivity on the retained image. It carries the identity to $JJ^\dagger$, not the full fine identity. Source-dependent families require

$$\partial_s(J^\dagger H_f J) = J^\dagger(\partial_s H_f)J + (\partial_s J^\dagger)H_f J + J^\dagger H_f(\partial_s J).$$

State and discarded-normalization derivatives must also be retained. The finite polynomial identities are Lean-checked; evolution follows by functional calculus. Coherent renewal must hold on the retained state. These relations connect geometry, native clocks, and source response for the supplied operators.

The capacity premises fix neither a unique gluing tensor nor a geometric measure. Absolute angular transfer, normalized connected patches, and metric-dependent reconnection amplitudes define concrete extensions. Each extension has a definite transfer operator whose phase and continuum behavior can be tested. The face-wise lock and the connected patch construction give different ensembles.

14.2 Coherent bonds, curvature, and physical calibration

The capacity and orientation representation admits an equivariant encoding

$$\mathcal{H}_{\mathrm{cap,o}} \cong \mathcal{F}(\mathbb{C}^4)\otimes\mathbb{C}^{105}, \qquad \rho_{\mathrm{cell}} = \frac{I_{16}}{16}\otimes\sigma.$$

Pair neighboring virtual fermionic ports in $(|10\rangle+|01\rangle)/\sqrt{2}$. Every cell retains the original 1680-state marginal, while each cut bond carries $\ln 2$ of conditional entropy. The residual entropy $S(\sigma)$ is local. The spatial bond state is selected; the local Gibbs weights also admit unentangled global states.

An absolute angular-transfer rule supplies the curvature normalization:

$$K_h(\theta) = e^{-\theta\ln 2/(2\pi)} I_2, \qquad K_h(2\pi) = \rho_h = I_2/2.$$

The same tensor gives the replica trace 2^{1-n} and hinge weight $\exp[(\ln 2/2\pi)N_h\delta_h]$. With $A_h = N_h A_{\mathrm{ch}}$, comparison with the Regge action gives

$$\boxed{\frac{\hbar G_0}{c^3} = \frac{A_{\mathrm{ch}}}{4\ln 2}.}$$

The full-turn normalization is microscopic action data. Normalized cut entropy alone would leave this bulk coefficient undetermined.

Physical area requires an event model. Assume Poisson placement in invariant four-volume, mean event volume v_{ev}, and a longest-chain clock between interaction endpoints. In four flat dimensions,

$$V_4(\tau) = \frac{\pi}{24}(c\tau)^4, \qquad L_{\mathrm{chain}} \sim m_4(V_4/v_{\mathrm{ev}})^{1/4}.$$

Matching its asymptotic cadence to $1/\tau_*$ gives

$$\frac{v_{\mathrm{ev}}}{L_*^4} = \frac{\pi}{24}m_4^4.$$

This event law is additional to renewal of the capacity labels. The chain rule is variational and presupposes causal geometry; strong confinement can alter the cadence.

Assigning one mean event per paired cell per proper tick and using a regular mean tetrahedral area–volume relation gives

$$V_{\mathrm{site}} = \frac{v_{\mathrm{ev}}}{2L_*}, \qquad a = (6\sqrt{2}V_{\mathrm{site}})^{1/3}, \qquad A_{\mathrm{ch}} = \frac{\sqrt{3}}{4}a^2.$$

Hence

$$\boxed{\frac{G_0}{G_*} = 0.10551926870131369\, m_4^{8/3}.}$$

At the illustrative finite-size estimate $m_4 = 2.35$,

$$\frac{v_{\mathrm{ev}}}{L_*^4} = 3.99218, \quad \frac{a}{L_*} = 2.56812, \quad \frac{A_{\mathrm{ch}}}{L_*^2} = 2.85582, \quad \frac{G_0}{G_*} = 1.03002.$$

A compatible mixed quantum volume state exists; its mean was imposed, so its dynamical selection remains open. Exact equality would require $m_4 = 2.32408$, a target implied by the calibration. The reference G_*, the angular action's bare G_0, and the interacting coefficient G_R have distinct definitions.

14.3 Connected primitive geometry and its history measure

For Euclidean causal dynamical triangulations, the spatial and mixed triangle areas obey

$$A_{\mathrm{SL}} = \frac{\sqrt{3}}{4}a^2, \qquad A_{\mathrm{mix}} = \frac{\sqrt{4\alpha - 1}}{4}a^2, \qquad \alpha = \frac{1 + 3(n/m)^2}{4}$$

when they contain m, n binary channels. A four-port patch has an even number $4V - 2E$ of external ports. Mixed tetrahedral bundles $(1, n, n, n)$ therefore require odd n. The first nonisotropic choice is $n = 3$, giving $\alpha = 7$. The required coarse intertwiner spaces are

$$\mathrm{Inv}(V_3^{\otimes 4}), \quad \mathrm{Inv}(V_3 \otimes V_9^{\otimes 3}), \quad \mathrm{Inv}(V_9^{\otimes 4}),$$

of dimensions $7, 7, 19$, respectively. Spin nine is the maximal fusion of three primitive facets.

A positive pair parent selects a 13-dimensional subspace of the 49-dimensional two-cell intertwiner space. Four cells in a chain, with two such pair restrictions, map onto the mixed seven-dimensional space. Six cells in a cycle, with three pair restrictions, map onto the nineteen-dimensional space. Numerical rank and coherent-contraction checks verify these maps, with held-out residual 3.29×10^{-15}. They are connected spin networks with coarse polyhedral observables [149]. These patches encode coarse polyhedra and need not be literal tetrahedral subdivisions. The tested six- and eight-tetrahedron balls reach rank thirteen.

For either surjective patch map R_e, full Gram normalization gives

$$I_e = R_e^\dagger (R_e R_e^\dagger)^{-1/2}, \qquad I_e^\dagger I_e = I.$$

Inside the paired ground space, $\Delta(P_{\mathrm{pair}} - I_e I_e^\dagger)$, $\Delta > 0$, is a positive parent for the code. The isometry errors are below 2.1×10^{-14}. Transported coarse observables define the encoded geometry; they need not equal sums of unmodified fine volume operators. Extending the pair parent indiscriminately over every graph edge gives soft modes with a path gap bounded by $O(n^{-2})$, so that interaction alone does not supply relativistic propagation.

Choose normalized coherent boundary packets and simplex factors

$$w_\sigma = \exp\left[-b \sum_{h \subset \sigma} N_h \theta_{h\sigma} - \lambda V_\sigma\right], \qquad b = \frac{\ln 2}{2\pi}.$$

Shared packets contract to one. The boundary half-turn factor

$$B_h(\theta) = \rho_h^{-1/2} K_h(\theta) = e^{b(\pi-\theta)} I_2$$

sews to the full-turn factor. Every allowed coarse history thus has

$$W(\mathcal{T}) = \exp\left[b \sum_{h \in \mathrm{int}} N_h \delta_h + b \sum_{h \in \partial} N_h \psi_h - \lambda \sum_{\sigma} V_\sigma \right].$$

On 1,920 causal slabs the maximum log-weight discrepancy is 1.46×10^{-13}. Raw patch contractions would change this measure through shape-dependent norms. The normalized construction is a selected Euclidean realization, distinct from the standard EPRL vertex and from a Lorentzian representation lift.

The host counting prescription is

$$T(g_1, g_0) = \sum_{[\mathcal{T}] \in \mathcal{C}(g_1, g_0)} \frac{z_{41}^{N_{41}} z_{32}^{N_{32}}}{C(\mathcal{T})} W(\mathcal{T}).$$

Here C is the automorphism factor; fugacities or fixed volumes specify the ensemble. The two-step and orthonormal kernels are

$$T^{(2)}(g_2, g_0) = \sum_{g_1} C(g_1) T(g_2, g_1) T(g_1, g_0),$$
$$\widetilde{T}(g', g) = \sqrt{C(g') C(g)}\, T(g', g).$$

At a finite symmetric cutoff its square is positive semidefinite. This does not make each one-step eigenvalue positive. The encoded transfer $J \widetilde{T} J^\dagger$ preserves composition. Sampling move sequences add no physical history multiplicity.

14.4 Metric counts, ordering, and physical states

Six directional block cuts reconstruct a spatial metric. For normals e_i and $(e_i + e_j)/\sqrt{2}$, let $r_A = A_A / A_0$. Then

$$\mathcal{C}_{ii} = r_i^2, \qquad \mathcal{C}_{ij} = r_{ij,+}^2 - \frac{r_i^2 + r_j^2}{2}, \qquad \boxed{g = \sqrt{\det \mathcal{C}}\, \mathcal{C}^{-1}}$$

on $\mathcal{C} > 0$. These are coarse cuts, not additional faces of one tetrahedron. Let $q_A = r_A - 1$ and $J_{aA} = \partial g_a / \partial q_A$. The DeWitt mobility in count coordinates is

$$K = J^{-1} D(g) J^{-T}, \qquad D^{ab} = \frac{\mathrm{tr}(E_a g E_b g) - \frac{1}{2}\mathrm{tr}(g E_a)\mathrm{tr}(g E_b)}{\sqrt{\det g}},$$

where E_a is an orthonormal symmetric-matrix basis. In this subsection use $\hbar = c = 1$. The selected Hamiltonian has

$$H_K = \frac{16\pi G_0}{V_{\mathrm{ref}}} p_q^T K p_q, \qquad H_R = -\frac{V_{\mathrm{ref}}}{16\pi G_0} \sqrt{\det g}\, R[g].$$

For M binary reconnection switches, $S^+|n\rangle = \sqrt{(n+1)(M-n)}|n+1\rangle$. Dividing by this multiplicity implements unit count shifts; products give the correlated moves required by K. The microscopic amplitudes implement the supplied metric action.

The angular Regge Hessian approaches the Einstein quadratic operator. Four gauge directions and one negative off-shell conformal direction leave two physical Lorentzian TT modes. The same conserved static source gives $h_{44} = -8\pi G_0 \rho / k^2$, hence $\nabla^2 \Phi = 4\pi G_0 \rho$. Two-momentum extrapolations agree with the angular coefficient to better than 10^{-7} relatively on the tested backgrounds. The nonlinear mobility also reproduces Kasner evolution with exponents $(-1/3, 2/3, 2/3)$. These are saddle and finite-operator tests, without an interacting phase determination.

The induced kinematic density and transported metric Weyl ordering are

$$\mu(q) \propto \frac{\prod_A r_A}{(\det \mathcal{C})^{5/4}}, \qquad H_{W,\mathrm{count}} = H_{\mathrm{LB,count}} - c_K \frac{21}{16\sqrt{\det g}}, \qquad c_K = \frac{16\pi G_0}{V_{\mathrm{ref}}}.$$

The ordering term is proportional to $\hbar^2$ when units are restored. With continuous metric configurations at fixed spatial regulator, the relevant Weyl commutators exactly quantize the regulated classical brackets. This preserves their regulator defects. On a three-cell probe, increasing mean counts from 100 to 800 lowers the count-current discrepancy from 1.04584% to 0.0163828% with second-order convergence. Its separate spatial Lie-derivative error is 8.5389%.

A small reflecting count box has no zero Weyl-constraint state. The unbounded homogeneous metric domain admits a different construction. Write

$$g = e^{2\alpha_{\mathrm{vol}}} h, \quad \det h = 1, \quad d\mu \propto e^{9\alpha_{\mathrm{vol}}} d\alpha_{\mathrm{vol}}\, d\mathrm{vol}_{\mathrm{sh}}, \quad h \in SL(3,\mathbb{R})/SO(3).$$

After the half-density transform, the congruence with $B = e^{3\alpha_{\mathrm{vol}}/2}$ gives

$$\widetilde{C} = \frac{24}{c_K} B C_{\mathrm{flat}} B = \partial^2_{\alpha_{\mathrm{vol}}} - 24\Delta_{\mathrm{sh}} - \frac{81}{2}.$$

For a self-adjoint shape Laplacian with spectral value $\lambda > 27/16$, put $\omega_\lambda = \sqrt{24\lambda - 81/2}$. The spectral form

$$\widetilde{\eta}(f,g) = \sum_{s=\pm} \int_{\lambda > 27/16} \frac{d\nu(\lambda,\sigma)}{2\omega_\lambda}\, \overline{\widehat{f}(s\omega_\lambda,\lambda,\sigma)}\, \widehat{g}(s\omega_\lambda,\lambda,\sigma)$$

is positive and annihilates $\widetilde{C}$. The original constraint has $\eta_C(f,g) = \widetilde{\eta}(Bf, Bg)$ on compatible test domains away from threshold. Quotienting null vectors gives a positive homogeneous physical inner product. Domain and rescaling choices matter [150]. A generalized shape mode at $\lambda = 5/2$ gives second-order residual convergence in the original area stencil. Inhomogeneous local constraints, metricity boundaries, and common physical-state refinement remain open.

14.5 Histories, constrained response, and an auxiliary clock

Use $\hbar = c = 1$ in this subsection. The finite Hermitian count operator has an ordered-jump history expansion with factors $-H_{yx}$. A triangle product -0.254534167364 in the 729-state test is invariant under diagonal rephasing, so these histories cannot all have positive weights. Even isolating the negative scale direction leaves signed cycles in the tested shape stencil. The canonical kernel and the positive covariant weight of Section 14.3 are distinct specified constructions.

Static density requires the constrained metric components. For a Gaussian contour $h = Ry$, both source and observable rotate, giving covariance $R|\lambda|^{-1}R^T = K^{-1}$. The checked static response gives $G/G_0 = 1$. Deleting the negative direction gives $4/3$; replacing it by its absolute eigenvalue without rotating the source gives $5/3$; keeping only spatial TT waves gives zero.

A conditional nonlinear integration follows for $g_{\mu\nu} = \Omega^2 \widehat{g}_{\mu\nu}$ at $\Lambda = 0$. With $c_R = 1/(16\pi G_0)$, the appropriate boundary treatment, and proper-length source j,

$$S_E = -c_R \int \sqrt{\widehat{g}}\, [\Omega^2 \widehat{R} + 6(\widehat{\nabla}\Omega)^2] + \int \sqrt{\widehat{g}}\, j\Omega.$$

Fix the mean scale and set $\Omega = 1 + i\chi$, with zero-mean real χ. If P removes the constant mode, define

$$A = P(-6\widehat{\Delta} + \widehat{R})P, \qquad \eta = P(j - 2c_R \widehat{R}).$$

For $A > 0$ and flat contour measure, integration gives

$$S_{\mathrm{eff}} = -c_R \int \sqrt{\widehat{g}}\, \widehat{R} + \int \sqrt{\widehat{g}}\, j + \frac{\langle \eta, A^{-1}\eta \rangle}{4c_R} + \frac{1}{2}\log\det A$$

up to common normalization. A cosmological or fixed-volume term introduces Ω^4; the flat contour measure is not derived from finite count histories. Coordinate density itself cancels between a kernel $K_{xy} = T_{xy}/\sqrt{\mu_x \mu_y}$ and the closed-history integration measure. Inserting it again changes the model.

A supplied bounded positive lapse N also permits an auxiliary proper-clock construction. Let $H_N = N^{1/2} H_0 N^{1/2}$ and $\mathbb{H} = H_N + N P_\tau$, with self-adjoint count generator H_0 and ideal auxiliary momentum P_τ. At fixed momentum p,

$$U_t(p) = e^{-it(H_N + pN)}, \qquad \mathcal{T}_t = i U_t^\dagger \partial_p U_t\big|_0 = \int_0^t U_s^\dagger N U_s \, ds.$$

The fixed-reference-energy equation $(H_N - iN\partial_\tau)\Psi = E\Psi$ becomes

$$i\partial_\tau \chi = (H_0 - EN^{-1})\chi, \qquad \chi = N^{1/2}\Psi.$$

It is unitary on a self-adjoint domain and preserves the scalar flux $\Psi^\dagger N \Psi$. Volume turning points do not stop this clock; lapse zeros require a new inverse-lapse domain. This auxiliary construction is distinct from the retained native clock below. Multiple local clocks must satisfy the normal deformation algebra, including its spatial diffeomorphism term.

14.6 One finite geometry, orbital, and native-clock operator

Use geometric cochains for the retained position space and keep the internal $Q \cong \mathbb{C}^9$ factor. With coboundary d and positive cochain metrics M_p,

$$d_p^* = M_p^{-1} d_p^\dagger M_{p+1}, \qquad D_g = d + d^*, \qquad \Omega_g(m) = \sqrt{D_g^2 + m^2 I}$$

give a selected positive-energy Dirac–Kähler orbital completion [151]. In units $\hbar = c = L_* = 1$, the native input is

$$m = \epsilon_e = -\frac{3}{2} Z_e \log(1 - e^{-7 g_{\text{share,eff}}}) = 4.18545879558 \times 10^{-23}.$$

Harmonic modes have energy m, and $\partial_m \Omega_g = m(D_g^2 + m^2 I)^{-1/2}$. For positive lapse and mass-independent geometric coupling,

$$H(m) = \widehat{H}_g + \bigoplus_g N_g^{1/2} \Omega_g(m) N_g^{1/2}, \qquad \mathcal{T}(t) = i U_t^\dagger \partial_m U_t = \int_0^t U_s^\dagger (\partial_m H) U_s \, ds \succeq 0.$$

Real-time evolution is unitary. If the spatial symbol is $E = \sqrt{h^{ij} p_i p_j + m^2}$, then $d\tau/dt = Nm/E$ and the worldline action is $-m \int d\tau$. The clock has no uniform positive lower bound as the momentum cutoff grows.

A sufficient Euclidean construction starts from a supplied half-transfer A_0, positive duration operator N_0, and $K(m) = A_0^\dagger e^{-mN_0} A_0$. On its fixed support, $C_m = -\partial_m K \succeq 0$ and

$$\partial_m \left(-\frac{\log K}{\Delta t} \right) = \frac{1}{\Delta t} \int_0^\infty (K + sI)^{-1} C_m (K + sI)^{-1} \, ds \succeq 0.$$

Bounds $n_{\min} I \preceq N_0 \preceq n_{\max} I$ give the same bounds on the clock rate divided by Δt. Positive entries with assigned lengths alone are insufficient. The half-transfer and duration operator are additional data; replacing a duration distribution by its mean loses cumulants.

An explicit geometric transfer couples the five-tetrahedron boundary of a four-simplex to the boundary obtained by replacing one tetrahedron with four regular tetrahedra. These are different physical metrics, of cochain dimensions 30 and 44. Their four boundary-pair families contain $1920, 1992, 1992, 2064$ causal slabs at $\alpha = 7$. The angular sums give

$$A = \begin{pmatrix} 0.1169508311 & 0.3422031803 \\ 0.3422031803 & 1 \end{pmatrix}, \qquad K_g = A^\dagger A, \qquad H_g = -\frac{\log K_g}{\Delta t}, \quad \Delta t = 13.94846613.$$

The duration is a declared synchronization of two boundary-normal intervals. Raw stellar interpolation would shift the marked geometric ground energy by 0.131992, overwhelming the native gap. Separate polar normalization of the harmonic and excited sectors gives an isometric transport R that maps harmonics to harmonics. The coupled geometric operator is

$$\widehat{H}_g = \begin{pmatrix} (H_g)_{00} I & (H_g)_{01} R^\dagger \\ (H_g)_{10} R & (H_g)_{11} I \end{pmatrix}.$$

It acts identically on the internal marked factor. Its lower bound is the empty-sector ground energy; harmonic modes attain that bound with orbital energy m. At unit reference lapse this yields

$$\boxed{E_{\mathrm{marked},0}(m) - E_{\mathrm{vacuum},0} = m.}$$

This protects the tiny gap algebraically. At control mass 0.23, its numerical error is 4.45×10^{-16}; clock differentiation and unitarity agree to 2.84×10^{-10} and 1.76×10^{-15}. A retained packet moves and changes geometry coherently.

For two descriptions of the same supplied coarse geometry, choose cochain maps P, R_c with $R_c P = I$, both commuting with d, and set $Q_f = I - P R_c$. The inverse fine metric

$$\boxed{M_f = R_c^\dagger M_c R_c + Q_f^\dagger M_a Q_f, \qquad M_a > 0}$$

makes the coarse image and complement orthogonal invariant subspaces. In orthonormal coordinates,

$$J^\dagger J = I, \quad D_f J = J D_c, \quad \Omega_f J = J \Omega_c, \quad (\partial_m \Omega_f) J = J (\partial_m \Omega_c).$$

Functional calculus also preserves evolution exactly. Two refinements $30 \to 44 \to 58$ preserve the two harmonic modes with no additional zero modes in the tested complement. The complete two-geometry lift $74 \to 102$ preserves generator, clock, evolution, and transported test sources to about 10^{-13} or better. The generally off-diagonal metric realizes a supplied operator; it preserves any unwanted coarse dispersion or species as well.

Regular-cell weights give

$$v_0 = \sqrt{5/6}, \qquad v_T = \sqrt{4/5}, \qquad v_3 = \sqrt{3/4}.$$

The tested inverse coarse-form metric restores the leading periodic cone, with a subleading transverse split. On tested closed periodic complexes, Dirac–Kähler cochains have the torus Betti counts $(1, 3, 3, 1)$, two Dirac tastes, and no additional lattice doublers. Flat-realization Voronoi–Delaunay Hodge weights give a common leading speed; regular-cell weights give each form sector its own cone. The dimension-five term vanishes on centrosymmetric complexes and in the mirror-symmetric ensemble, and its per-realization coefficient falls as $N^{-1/2}$, far below the Crab bound. Constant per-degree rescaling fails across different complexes. The refreshed lift is inert when fermions use the record-oriented frame; the complex's coherent spatial orientation persists under gluing. Selection of the physical Hodge metric, taste and chiral assignments, changing-boundary consistency, and radiative stability remain open.

The common low-curvature target is

$$\Gamma[g, \psi] = \frac{1}{16 \pi G_R} \int \sqrt{-g}(R - 2\Lambda) + \Gamma_m[g, \psi] + \Gamma_\partial + \sum_i c_i \mathcal{O}_i.$$

The finite constructions do not yet fix its renormalized coefficients or higher-curvature terms. A spacetime solution also requires initial and boundary conditions.

15. Dynamical-Lattice Tests

Capacity-decorated causal-dynamical-triangulation (CDT) simulations test the local incidence and its coupling to a fluctuating host. The original face-wise campaign, the normalized connected model, and continuous shape interpolations use different geometric measures. Their recorded results apply to their stated ensembles.

At matched volume near $N_{41} \simeq 10^5$, turning on the admissibility weighting changes the collision fraction from 0.654 to 0.077, while a shuffled-energy control remains at 0.736; the global host geometry remains stable with measured Hausdorff estimators around 3.55–3.59. Persistent closure-failure defects produce significant local capacity and coordination responses in two independent matched comparisons.

A separate conserved-transport experiment finds exact conservation and an approximately additive maintenance charge,

$$Q - Q(0) = (0.1440 \pm 0.0025)m,$$

with one shell-one source-to-field ratio across $m = 1, 2, 3, 6$,

$$\frac{\Delta f_1 - \Delta f_1(0)}{Q - Q(0)} = 4.9965 \pm 0.5244.$$

The dressing-subtracted profile stays positive through eight shells and fits $d^{-0.62\pm0.11}$; screening is consistent with zero, giving $\xi > 4.6$ steps. The source fit has a maximum relative residual 0.242, mainly at $m = 6$. The finite response therefore shows nonlinear saturation; its continuum $1/r$ limit and absolute junction to $8\pi G_*/c^2$ remain open.

The decorated continuum target requires an extended semiclassical phase with

$$d_H \to 4, \qquad N_3(t) \propto \cos^3(t/s_0), \qquad \lambda_{\mathrm{TT}}(k) = 1 - c_{\mathrm{TT}} k^2 a^2 + O(k^4).$$

The reference-host volume scaling does not establish these conditions. The original face-wise lock predicts a native point near $(\kappa_0, \Delta) = (2\ln 2, 0.01)$. Its bare-measure volume-pair test fails while the reference point passes, with seeds sampling extended and concentrated basins. This constrains the original bare measure; the decorated and connected hosts require their own phase tests. The frozen-area shape estimate $L_{\mathrm{block}} \simeq 3.4 L_*$ belongs to that branch, whose shape modes do not supply spatial TT gravitons. A continuous critical surface is a stronger optional cutoff-removal test.

15.1 Normalization and distinct geometric ensembles

A homogeneous blob fit with width $\widetilde{s}_0$, kinetic coefficient k_1, and n_{ev} physical events per counted simplex gives, under its Einstein minisuperspace interpretation and in geometric units,

$$\frac{G}{L_*^2} = \frac{\widetilde{s}_0^2}{3\sqrt{6}\, k_1} \sqrt{n_{\mathrm{ev}} \frac{v_{\mathrm{ev}}}{L_*^4}}.$$

The spacing, shape, and volume conventions cancel. The physical clock, event assignment, and identification of this fitted mode with universal gravity remain premises. Applying the paired calibration consistently to the earlier public pilot gives $G_{\mathrm{bulk}}/G_{\mathrm{cut}} = 1.5020714592$, independent of m_4. That mismatch is not removed by a more precise event-volume conversion.

The connected $\alpha = 7$ model has

$$\frac{a^2}{G_0} = \frac{16\ln 2}{\sqrt{3}} = 6.40301938055.$$

For

$$S_E = -(\kappa_0 + 6\Delta)N_0 + (\kappa_4 + 2\Delta)N_{41} + (\kappa_4 + \Delta)N_{32},$$

the curvature map is

$$(\kappa_0, \Delta, \kappa_4^{\mathrm{curv}}) = (2.20205311516,\ 0.326138328034,\ -0.304774619960).$$

The continuous interpolation at $\alpha = 11.312$ instead gives $(2.70257004, 0.43691236, -0.42026496)$. These differ from the original face-wise near-unit-asymmetry campaign. The curvature contribution to κ_4 is not its critical tuned value; volume constraints and fugacities remain ensemble data. Changing m_4 rescales a^2 and G_0 together, without changing their ratio.

With $r = N_{32}^{\mathrm{slab}}/N_{41}^{\mathrm{slab}}$, the trapezoidal volume-equivalent duration is

$$\frac{c\tau_{\mathrm{vol}}^{L,E}}{L_*} = \frac{a/L_*}{4\sqrt{2}}\left[\sqrt{8\alpha \pm 3} + r\sqrt{12\alpha \pm 7}\right],$$

where plus is Lorentzian. At $\alpha = 11.312$, the Lorentzian value is $4.38972153 + 5.42399632r$. A host interval is not one native tick by definition. Volume-equivalent duration, geodesic length, and the mass-derivative clock are separately calibrated observables. The exact normalized tensors permit direct vacuum sampling of their CDT weight; existing runs with different measures do not test that branch.

15.2 Native sources and the interacting force test

The native rest generator gives the selected proper-time attachment

$$T_e[\gamma] = e^{-\epsilon_e P_e \tau_\gamma/\tau_*}, \qquad U_e[\gamma] = e^{-i\epsilon_e P_e \tau_\gamma/\tau_*}.$$

The same energy supplies inertia and the stationary lapse coupling. On all 1,920 $\alpha = 7$ slabs, transported normal geodesics have durations of roughly 6.54–6.99 ticks, and the native transfer-sewing error is 1.72×10^{-16}. Their free-line covariance is -0.00171521593106, giving a leading native connected logarithm of $-3.00472647 \times 10^{-48}$. These freely moving lines do not measure a static force. A fixed timelike edge per slice has geometry-independent length and gives zero connected rest-mass response.

For held, mass-independent source preparations, let L_1, L_2 be Euclidean proper lengths. In units $\hbar = c = 1$,

$$R = \frac{\langle e^{-m_1 L_1 - m_2 L_2}\rangle}{\langle e^{-m_1 L_1}\rangle\langle e^{-m_2 L_2}\rangle}, \qquad Q(r,T) = -\frac{\mathrm{Cov}(L_1, L_2)}{T}.$$

The mixed derivative of $\log R$ is the covariance. An Einstein infrared regime requires $Q \to -G_R/r$. The supplied Gaussian propagator gives the finite-duration normalization check

$$Q(r,T) = -\frac{2G}{\pi r}\left[\arctan(T/r) - \frac{r}{2T}\log(1 + T^2/r^2)\right].$$

This benchmark assumes the propagator; the interacting triangulation response remains to be measured.

Intrinsic face transport and a held-separation rule define the sources. If $\dot{D} = A_D + e\cdot v$, the minimum-norm holding velocity is $v = -A_D e$, timelike when $-N^2 + A_D^2 < 0$. An explicit path bounds distance above; a continuous slice field with $|\nabla f| \le 1$ gives $f(q) - f(p) \le d(p,q)$. All 403 sampled points of the four-interval diagnostic pass the separation test, without a continuous supremum or whole-ensemble coverage theorem.

Normalized initial marks and complete geometric support give $Z_{00} = 1$. Filtering out failed source preparations would change the ensemble, so incomplete coverage suppresses the reported response. Initial-mark covariance also needs separation from geometry-induced covariance. The saved forty-input $\alpha = 11.312$ diagnostic uses a roughly two-percent Lorentzian volume window. It has no accepted complete force curve, only about five effective samples in the saved action diagnostic, and sub-edge separations $r/a = 0.3, 0.5, 0.7$. No infrared G_R follows from those data. The exact two-state estimator check verifies normalization, sign, and coverage handling; a physical measurement still needs equilibrium, $a \ll r \ll L_{\mathrm{box}}$, and duration stability on the chosen microscopic branch.

15.3 Completed connected-host normalization test

The registration and pre-data amendment of 17 October 2026 fix the $\alpha = 7$ curvature couplings, volume prescription and estimator before sampling. Q2 compares $R = G_{\rm vol}/G_0$ with the band 0.80–1.25. The primary volume is $N_{41} = 32{,}000$, and only runs passing the registered blob and stationarity gates enter its verdict.

Run	N_{41} target	R	$\sigma_{\ln R}$	Role
A2	16,000	1.060	0.065	Reported only
A4	32,000	1.064	0.089	Sole eligible primary
A5	64,000	0.893	0.054	Reported only

All three listed runs have stationary de Sitter-like profiles. The registered Q2 verdict is AGREE on A4, whose approximate two-standard-error interval is 0.89–1.27. Eligibility fails for A1, A3 and A6 because of drift; their ratios are retained in the full paper's J.14. The separate Q1 start-independence test is UNDETERMINED, including a 6.1-standard-error difference in largest-slice fraction between A3 and A4. The ratio R is independent of m_4 and tests the finite-volume stiffness normalization.

A separately registered curvature-trace test gives the diagnostic $\gamma_{\rm trace} = 1.82 \pm 1.04$ against $G_0/a^2 = 0.1562$. Its interval is too broad to decide the hypothesis, and both streams fail the registered maximum-width gate. Its verdict is UNDETERMINED. Exploratory scalar correlators on the saved geometries do not yield a stable binding energy. Matter-response production is deferred pending observable calibration. The full account and byte-preserved evidence packet are identified in J.14 of the main manuscript.

16. Equilibrium Vacuum Normalization

At fixed regulator, let

$$Z_{\rm cap}[T] = \mathbb{E}\exp\left[-\beta \sum_c E_c\right]$$

and define the homogeneous bulk free energy per cell by $\mu_{\rm cap}$. The centered factor is

$$\boxed{\widehat{Z}_{\rm cap}[T] = e^{\beta \mu_{\rm cap} N_{\rm cell}(T)} Z_{\rm cap}[T].}$$

By construction,

$$-\lim_{N\to\infty} \frac{1}{\beta N} \ln \widehat{Z}^{\rm vac}_{\rm cap} = 0.$$

This identity is invariant under $E_c \mapsto E_c + C$, $\mu_{\rm cap} \mapsto \mu_{\rm cap} + C$. Therefore an arbitrary source-independent microscopic zero-point shift cannot reappear as a capacity-sector observable; only defect- and geometry-dependent differences survive.

The uncentered lattice control verifies the sign and coefficient of the predicted bulk displacement. Exact Euclidean Regge four-volumes fix the volume operator in the Euclidean weight,

$$\mathcal{V}_4 = a^4 \left[v_{41}(\alpha) N_{41} + v_{32}(\alpha) N_{32}\right],$$

with

$$v_{41} = \frac{\sqrt{8\alpha - 3}}{96}, \qquad v_{32} = \frac{\sqrt{12\alpha - 7}}{96}.$$

Hence a continuum volume density $\rho_{\rm vol,R}$ couples through

$$t_{41} = a^4 v_{41} \rho_{\rm vol,R}, \qquad t_{32} = a^4 v_{32} \rho_{\rm vol,R},$$

so the volume normalization is an operator identity. Physical Lorentzian durations use the plus-sign volumes in Section 15.1.

The exact result is zero extensive capacity free-energy density in the centered reference. Calling its mapped capacity contribution zero requires the stated volume operator. It does not fix the total gravitational integration constant. At fixed event count a factor e^{-CN} cancels from normalized weights for every C; that invariance cannot determine the coefficient conjugate to volume. A common substrate can also carry several effective operators.

Conditioning a homogeneous Poisson process on total event count N gives a regional count of variance $Nf(1-f)$ for volume fraction f; a small region approaches the Poisson law. Fixing a total count leaves these local fluctuations. A pointwise fixed volume form is a stronger variational premise. With conserved stress it gives the trace-free Einstein equation,

$$R_{\mu\nu} - \frac{1}{4} R g_{\mu\nu} = \frac{8\pi G}{c^4} \left(T_{\mu\nu} - \frac{1}{4} T g_{\mu\nu} \right),$$

whose integrability leaves a cosmological integration constant.

The empty-equilibrium cosmological branch separately adopts

$$\boxed{\rho_{\Lambda,i} = 0}.$$

A common-parent continuum limit and a physical boundary or matching condition that selects this zero are distinct requirements. Later source-clock release is the conditional nonequilibrium sector of Section 11.

For nonequilibrium fluctuations, the exact Gaussian relative-information functional is

$$\boxed{\Gamma_{\mathrm{rel}}(C \| C_0) = \frac{1}{2} \mathrm{Tr} \left[\mathcal{G} - I - \ln \mathcal{G} \right], \qquad \mathcal{G} = C_0^{-1/2} C C_0^{-1/2},}$$

which is nonnegative and vanishes only at $C = C_0$. It supplies a controlled target class for positive state-dependent energy without reinstating a source-independent vacuum constant.

17. Conditional Gauge and Particle Extensions

The charged-lepton shell model is specified in Section 6, including its encoding and pole-mass identification. The remaining gauge construction is conditional.

A persistent open three-route record is a reversible qutrit channel. Removing the unphysical common channel phase gives

$$U(3)/U(1) = PU(3), \qquad \mathrm{Lie}[PU(3)] = \mathfrak{su}(3),$$

with eight traceless endpoint generators and an adjoint Wilson form. In the minimal identity-kernel completion, primitive $j_0 = 3/2$ recoupling fixes the octet transfer

$$\boxed{t_8 = \frac{13}{14}}.$$

This is a projective pure-gauge route construction, not yet QCD. A global $SU(3)$ lift, chiral quark matter, hypercharge, a chiral gauge-covariant fermion action, continuum-scheme matching, confinement, and hadron observables remain open. The existing $SU(2)$ primitive spin action is likewise not yet a derivation of the electroweak matter assignments.

18. Closure Status

Result	Grade	Reason
$7+9$ split and 1680 states	Closed in selected vertex	Oriented matching, equal-weight reversible complement, vector response, single-copy routing and orientation.
η_* and $g_{\mathrm{share,eff}}$	Closed finite calculation	Exact spectrum and analytic unique global maximum of normalized evidence.
Renewal and finite arrow	One-tick H-theorem closed; macroscopic arrow conditional	Faithful renewal fixes replacement and gives an exact one-tick H-theorem on the closure register; the persistent sector evolves unitarily. Record growth, the record root from exact recoverability and finite total capacity, and conditional typicality give the macroscopic arrow.
Electron cadence and L_*	Fixed within selected transfer and calibration	Marked generator sets the gap; electron calibrates τ_*. Finite orbital transport preserves it; a universal continuum cone remains open.
Seven-channel product minimum	Exact conditional bound	Fixed marginals, unchanged marked vertex, and specified positive-generator class.
Charged-lepton ladder	Selected pole-mass model	Six-label encoding, recurrence, common Gaussian overlap, and species identification. Shell multiplets and radiatively stable physical pole masses remain open.
G_*, G_0, and G_R	Distinct reference, bare, and infrared coefficients	Canonical static matching sets $G = G_*$. Angular transfer fixes $G_0 = A_{\mathrm{ch}} c^3/(4\hbar \ln 2)$. Universal interacting G_R remains a test.
Source map and static Einstein branch	Closed within canonical graph and matched parent	Green matching and Einstein constraints give Newton, no slip and PPN. Source-aware Gaussian reduction retains contact and determinant terms.
Finite-loop stiffness	Conditional	Return operator needs an independent microscopic audit; it cancels from κ/γ.
a_0, spherical RAR, transverse lensing	Conditional EFT equations; RAR derived from the horizon-thermal tilt mode	Tilt-mode identification, the Unruh reading that fixes the lever, projector and metric vertex are premises.
Resolved galactic response	Director action fixed; empirical tests mixed	Positive targets and angular-preserving radial updates exist. Director tests pass; three conditional live-stock realizations retain tilt for eight Gyr. Common dynamics, convergence and formation remain open.
Transport	Specified minimal completion	$D/\tau_0 = c^2$, $\tau_0^{-1} = H_0$. Quantum FDT survives reconstruction; Markov closure and initial transients require separate conditions.
Stock selection	Conditional effective dynamics	Bose relative-entropy contacts in the zero-temperature limit of detailed balance descend to the budget-capped exponential law; spherical runs reach it to 0.01–0.05 dex. Occupation identification, contact rate and exchange operator remain open.
Cluster stock	Fixed cohort budget; record identification conditional	$k_C = 5.333\, e^{-\eta_*}$; stock outside record-dominated supports keeps $1 + k_C = 6.18$; observed-cluster runs keep R_{500} within 0.1%. Assembly, member galaxies, mergers and lensing segregation remain open.
Formation and abundance	Specified map and ledger; vertex open	Bare pre-release ratio $1/\epsilon$ becomes 5.333 in the formation mass convention. Release depletes present dust. The specified labelled transfer forbids late recruitment.

Result	Grade	Reason
Release and clock dynamics	Exact identities inside a conditional model	Ensemble-clock exchange fixes perturbations. Composition-preserving allocation gives exponential depletion, valid only before local exhaustion; physical matching remains conditional.
Mixing and cosmological fit	Direct linear and exponential fits	The standard linear-depletion fit gives $\Delta\chi^2 = +0.85$ with one fewer parameter; exponential allocation gives -0.5152 with massless neutrinos, linear lensing and a low-redshift offset. The release history supplies the mixing normalization; the frozen history with a feedback-resolved retention gate gives $+1.07$ with two fewer parameters.
Clock and radial capacity	Exact Einstein reductions; observable extension conditional	$q_\tau = N^2$ obeys the static lapse equation; $q_R = 1 - 2GM_{\mathrm{MS}}/(c^2R)$. They agree in normalized static vacuum, differ inside matter. Generic relational capacity and boundary dynamics remain open.
Retained quantum sector	Finite reconstruction and infinite-volume algebra	Polar transport, controls and records give Born probabilities and instruments. Capacity bridges survive gauge projection; bounded interactions give quasi-local dynamics and GNS representations. Physical continuum state remains open.
Coherent bonds and angular action	Exact within selected spatial completion	Original cell marginal, $\ln 2$ cuts, full-turn normalization and Regge coefficient agree. The state and angular rule are physical inputs.
Event calibration	Conditional physical normalization	Poisson placement, longest-chain clock and paired regular mean cells give $G_0/G_* = 0.1055193 m_4^{8/3}$. Asymptotic m_4 and dynamical volume selection remain open.
Connected primitive host	Finite construction and rank checks	Normalized connected spin networks reproduce CDT weights at $\alpha = 7$. Coarse polyhedral geometry requires no literal tetrahedral subdivision of each patch.
Metric dynamics and ordering	Selected action; free TT Fock theory	Six cuts reconstruct the metric; DeWitt mobility and Weyl ordering reproduce the tested Hessian and brackets. Gaussian TT reflection positivity gives two graviton helicities; interacting phase remains open.
Physical constraint states	Homogeneous construction	Positive spectral form on a specified unbounded domain; inhomogeneous constraints and common physical refinement remain open.
Joint orbital and native clock	One finite coupled operator	Polar harmonic transport preserves the native gap. Inverse cochain metrics intertwine generator, clock, evolution and encoded sources through refinement.
Propagation and interaction	Finite controls; universal phase open	Record-oriented fermions are lift-invariant. Flat-realization Hodge weights give the tested common cone; regular-cell weights split it. Physical metric selection and infrared G_R remain open.
Coherent renewal	Exact compatible realization	Background-wide replacement and a common renewal frame preserve retained coherence. Distinct event counts and full refoliation require a common export construction.
Vacuum centering	Exact reference normalization	Removes capacity bulk free energy; it does not select the gravitational integration constant.

19. Sharp Falsifiers

The linked structure makes failures localizable within each stated model and matching prescription.

- **Static core:** a microscopic calculation that invalidates the $7 + 9$ incidence, exact closure spectrum, source theorem, or electron transfer gap would break the coefficient chain. Because G_* and both charged-lepton corrections share ζ_*, failure of the marked vertex would move them together.
- **Galactic branch:** the spherical RAR target must follow

$$g_{\rm obs} = g_{\rm bar}/[1 - e^{-\sqrt{g_{\rm bar}/a_0}}],$$

  while resolved nonspherical maps must satisfy the conservative field equation. The director completion carries two registered tests (SHA-256 `5a72f3e1a3ae0b65`...). Old-disk vertical dispersions of the DiskMass galaxies, measured apart from the young cold layer, reject it if the median ratio of measured to predicted dispersion falls outside 0.8–1.2; QUMOND predicts about 15% higher values. The persistent tilted stellar debris already challenges the aligned equilibrium solution. A disk-modeled stream population at 20–50 kpc with mean potential flattening below 0.85 or intrinsic scatter above 0.1 rejects that branch's nearly round excess, $q_\Phi = 0.95$–0.99. The live tilted-stock candidate is an additional conditional dynamical state; its population frequency and assembly history remain to be predicted. Persistent low-acceleration gravitational slip or a required Solar anomalous quadrupole would reject the transverse completion.
- **Clusters and groups:** relaxed clusters must hold total-to-baryon ratios near $1 + k_C = 6.18$ outside the central galaxy's support. Isolated X-ray ellipticals must follow the galactic law out to the radius where their hot gas outweighs their stars. In groups the excess over the law follows the stock not bound to the central galaxy, which reaches the core.
- **Hubble constant from galaxies:** $a_0 = \epsilon c H_0$ makes $H_0 = a_0/\epsilon c$ a galactic measurement. The standard linear-depletion fit gives $H_0 = 68.458\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, which requires $a_0 = 1.25005 \times 10^{-10}\,\mathrm{m\,s^{-2}}$; the Cepheid value 73.04 [60] requires $1.334 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. Only absolute distances carry this information. The 44 SPARC galaxies with such distances [53] give $a_0 = (1.21 \pm 0.19) \times 10^{-10}\,\mathrm{m\,s^{-2}}$, or $H_0 = 66 \pm 11$; a 2% determination of a_0 separates the two values at 3σ.
- **Redshift evolution:** the branch predicts $a_0(z) = \epsilon c H(z)$. Controlled high-redshift galaxies showing no corresponding evolution would reject the horizon lever. The electromagnetic coupling carries no horizon factor, so α stays fixed while a_0 follows $H(z)$. At $z = 1.15$, where a_0 is predicted to exceed its local value by a factor 1.95, quasar absorption gives $\Delta\alpha/\alpha = 1.3 \pm 1.3_{\rm stat} \pm 0.4_{\rm sys}$ parts per million [159].
- **Saturated cosmology:** with formation and mixing histories measured independently, the exponential allocation must fit the CMB, BAO and supernova data. Its weighted background and geodesic perturbations use the same formation convention.
- **Release/dark energy:** intrinsic vacuum stress has $p_\Lambda = -\rho_\Lambda$, while its separately conserved effective-fluid reading approaches $w_{\rm eff} = -1$ from below during accumulation. Decreasing physical vacuum density, or $w_{\rm eff} > -1$ for this same component after accounting for matter transfer, rejects the extension. A conserved-CDM w_0, w_a fit is a different reconstruction. Independent gates and kernels must reproduce the required release integral; substantial pre-gate identity loss or a local propagating determinant mode also rejects the model.
- **Strong field:** evidence that the proposed capacity saturation boundary cannot be reconciled with the GR exterior would reject the bounded-domain interpretation, while leaving the exact spherical Einstein reduction itself untouched.

- **Geometry and matter:** one chosen microscopic measure must give a massless TT sector, a common matter cone, and the same infrared Newton coefficient in static sources and curvature response. A volume profile alone cannot establish these properties. Loss of unmeasured position coherence, an unprotected native rest gap, or a persistent species-dependent cone rejects the corresponding matter completion.
- **Constraints and force:** count refinement and spatial refinement must jointly remove their distinct bracket defects on a common physical domain. Held-source measurements need complete support, equilibrium and an infrared separation window; a stable failure of $Q(r,T) \to -G_R/r$ there rejects that interacting branch.

20. Relation to Other Approaches

The proposal combines a finite capacity and record factor with explicit geometric hosts. The connected construction uses a normalized CDT history measure; the metric-Regge and spin-foam witnesses supply separate semiclassical checks. Capacity labels remain physical degrees of freedom distinct from geometric magnetic indices. Compared with the Λ cold-dark-matter model (ΛCDM), the phenomenological extension replaces particle dark matter with a carrier-resolved response and a conditional saturated early phase. Its inherited phase-cell and thermal matching premises specify the exponential MOND-like response. The ordinary longitudinal branch has an Einstein metric parent and introduces no independent scalar graviton.

Group field theory shares the tetrahedral quantum and the condensate reading of the vacuum [68, 70, 71, 69]. Continuum information-theoretic results recover the Einstein limit from horizon relative entropy [29, 30] and from a geometric relative-entropy action [32, 33]. None of these supplies the finite cell or its coefficients, which the present construction fixes.

Many-Pasts is likewise not many-worlds or a collapse theory. It retains one realized present, standard quantum dynamics, and ordinary Born/no-signaling predictions, while interpreting the compatible decoherent pasts as the history support of that present.

21. Conclusion

The selected vertex and clock give

$$V_1 \longrightarrow j_0 = \tfrac{3}{2} \longrightarrow 7+9 \longrightarrow 2P(7,4) = 1680,$$
$$\eta_* = 0.0298668\ldots, \qquad g_{\text{share,eff}} = 7.419800\ldots,$$
$$\text{faithful renewal} + \text{selected marked generator} + \text{electron} \longrightarrow \tau_*,$$
$$L_* = c\tau_* = 1.6162537 \times 10^{-35}\ \text{m}, \qquad G_* = \frac{c^3 L_*^2}{\hbar}.$$

A compatible coherent bond state and absolute angular transfer give

$$\frac{\hbar G_0}{c^3} = \frac{A_{\text{ch}}}{4\ln 2}, \qquad \frac{G_0}{G_*} = 0.1055192687\, m_4^{8/3}$$

under the Poisson, longest-chain, and paired-cell calibration. The near-unit ratio at $m_4 = 2.35$ is a conditional bare normalization. Equality with the physical interacting coefficient requires one common geometric state and infrared limit.

Connected patches supply normalized causal histories, coarse areas reconstruct a metric, and selected DeWitt dynamics admits a positive homogeneous constraint inner product. Isometric reduction preserves the encoded geometry–orbital operator, native gap, clock, and source response. The inhomogeneous physical domain, universal cone, coherent export across event counts, and interacting Einstein phase remain open.

The matched ordinary branch gives the Einstein constraints and nonlinear clock-capacity equation. Radial Misner–Sharp capacity agrees with clock capacity in normalized static vacuum and differs inside matter. Generic relational observables and saturation dynamics need specification. The shell model correlates lepton ratios; pole-mass realization and radiative stability remain tests.

The director EFT passes development and two old-disk comparisons. Positive stock and radial updates preserving angular information supply conditional realizations; the live tilted-stock candidate retains substantial tilt for eight Gyr. Its common variational dynamics and formation remain open. Ensemble-clock cosmology gives positive internal allocation, exponential depletion, and a local exhaustion boundary. Direct fits of linear depletion and exponential allocation give $\Delta\chi^2 = +0.85$ and -0.5152 against ΛCDM in the likelihoods of Section 11.

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